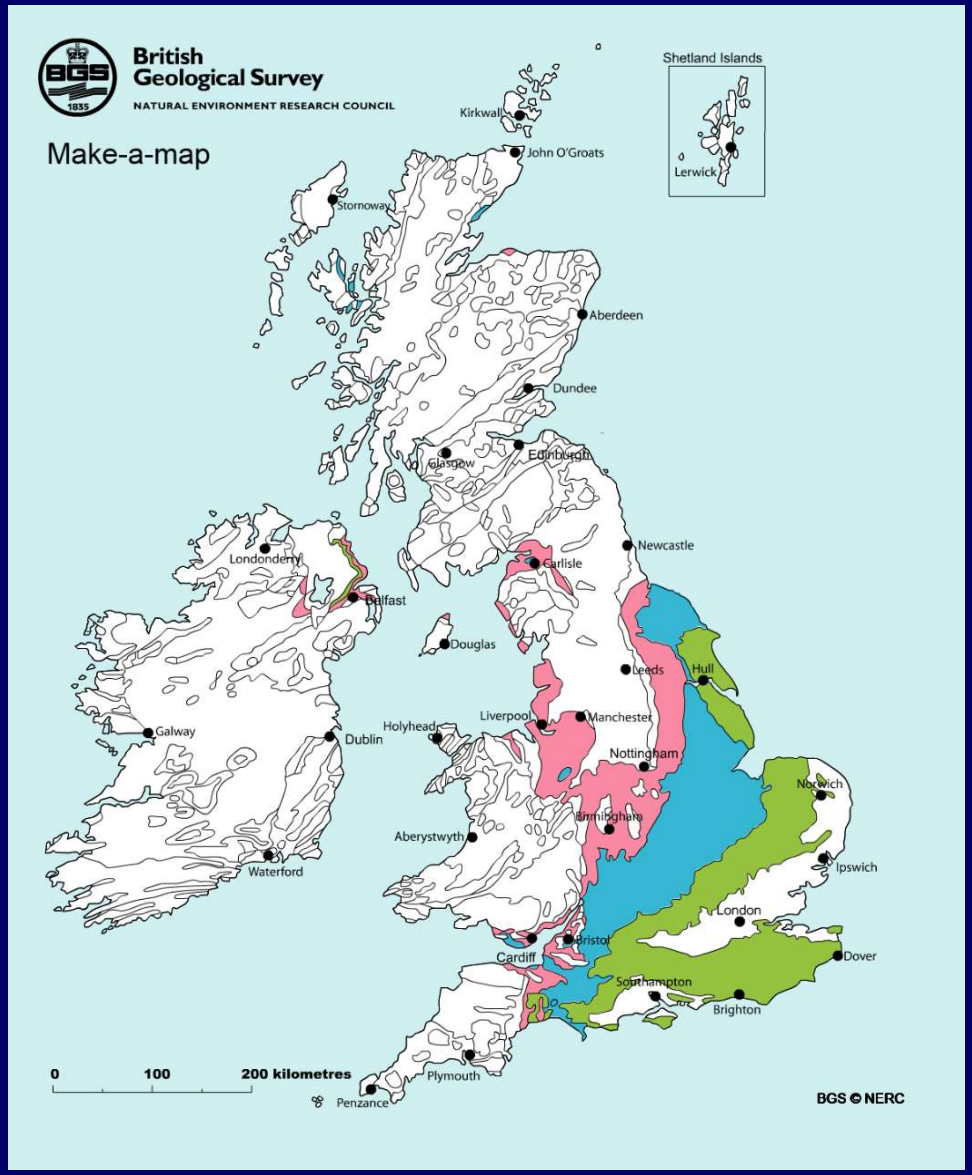
MANCHESTER 1824
The University of Manchester

Attendance: 281309

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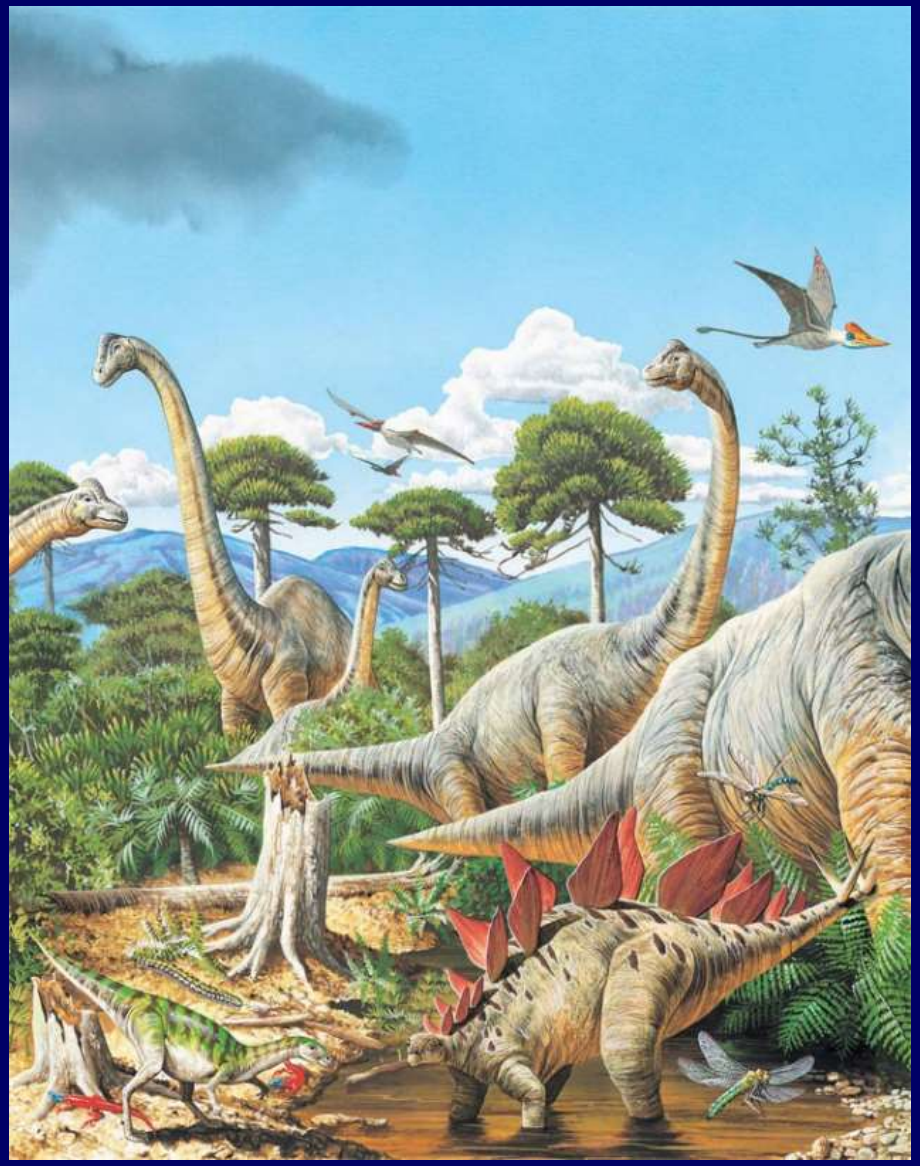
How to Build a Planet

Class 9
The Mesozoic Era: Legacy of a Supercontinent (252-66 Ma)



Distribution of Mesozoic age sedimentary rocks in the UK

What the Mesozoic is all about?

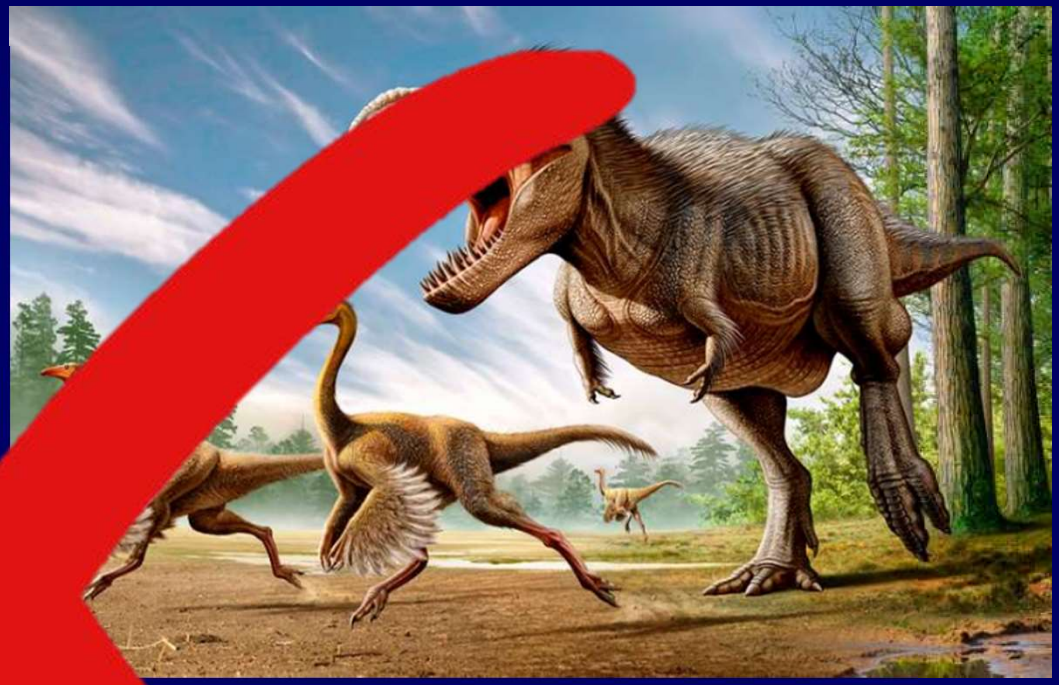






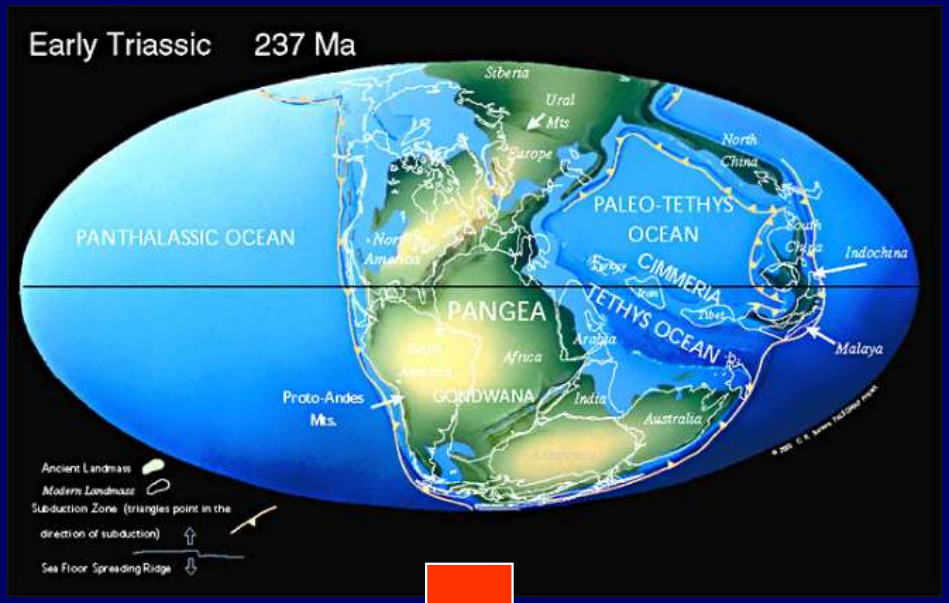
What the Mesozoic is all about: Not this!



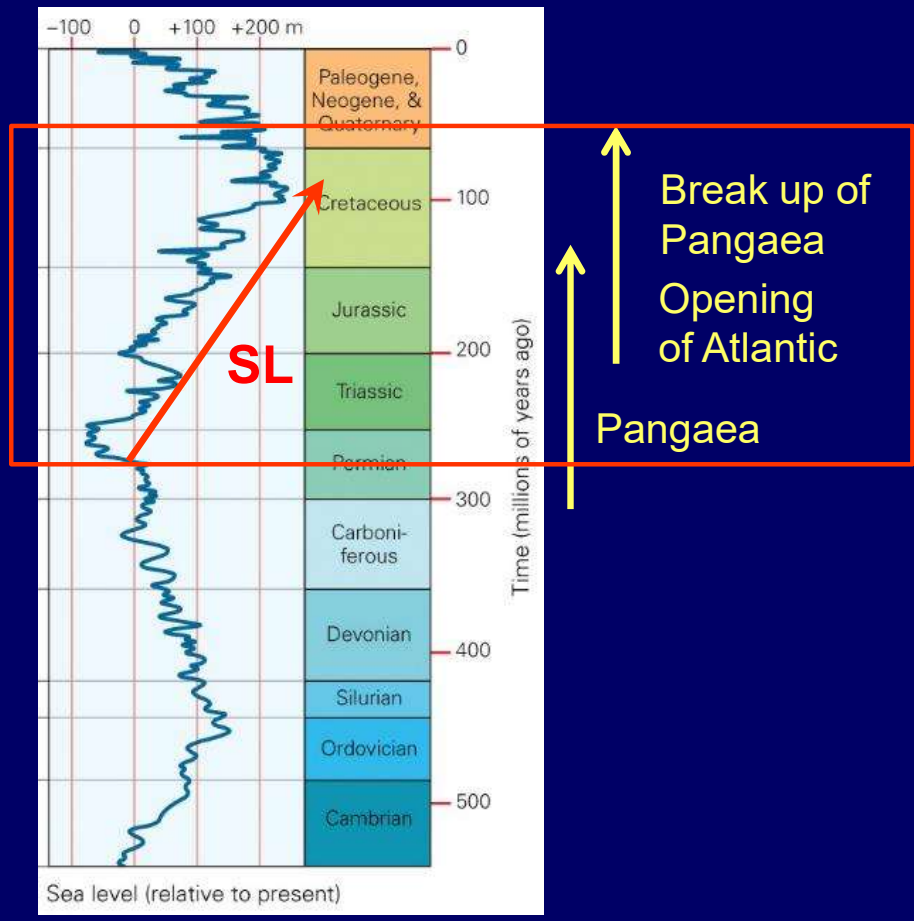




Changes in Mesozoic palaeogeography and sea level







Effect of break up of Pangaea and dispersal of continents:

- climate and ocean circulation patterns
- volcanism and erosion
- greenhouse crises → mass extinctions and evolutionary innovations
- sea level rose to >200m above present

About the Mesozoic Era: ‘Middle life’

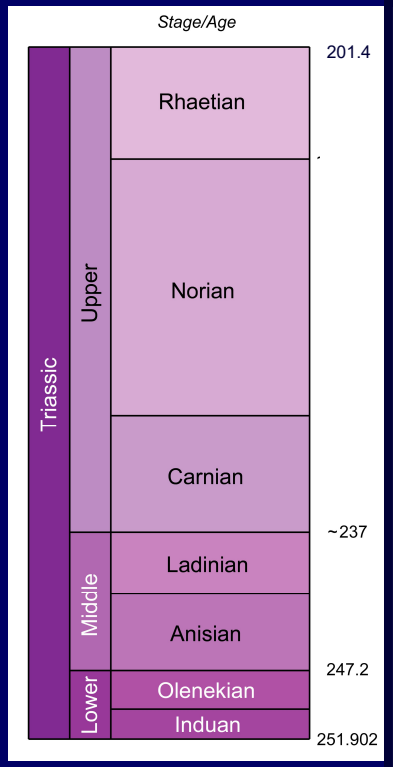


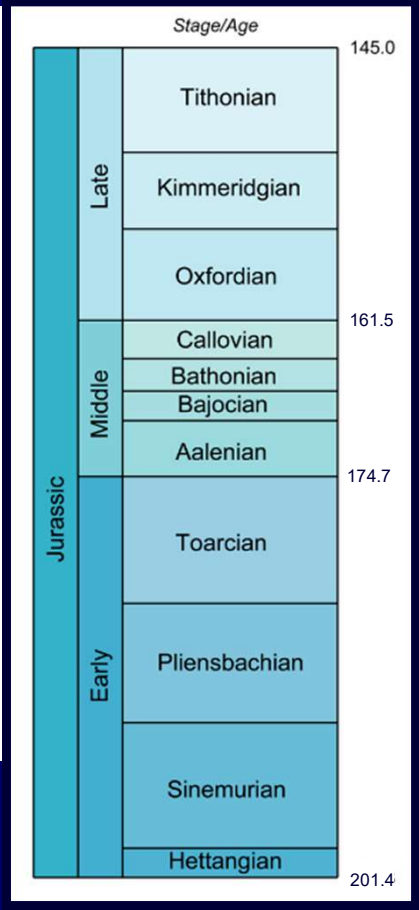
Red Triassic sandstone

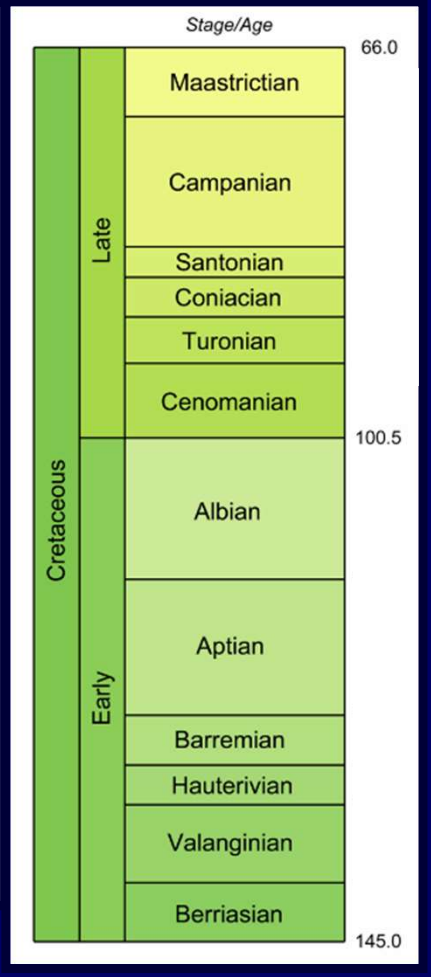


Triassic evaporites

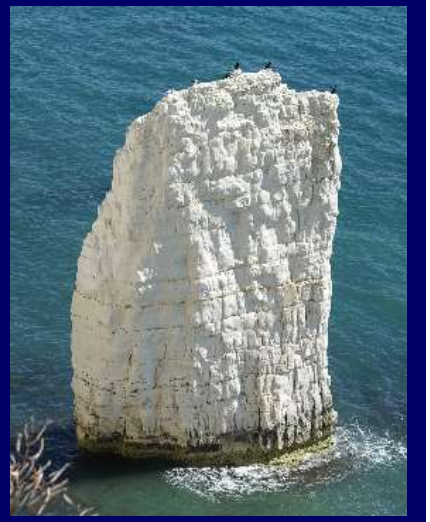
252 Ma







66 Ma



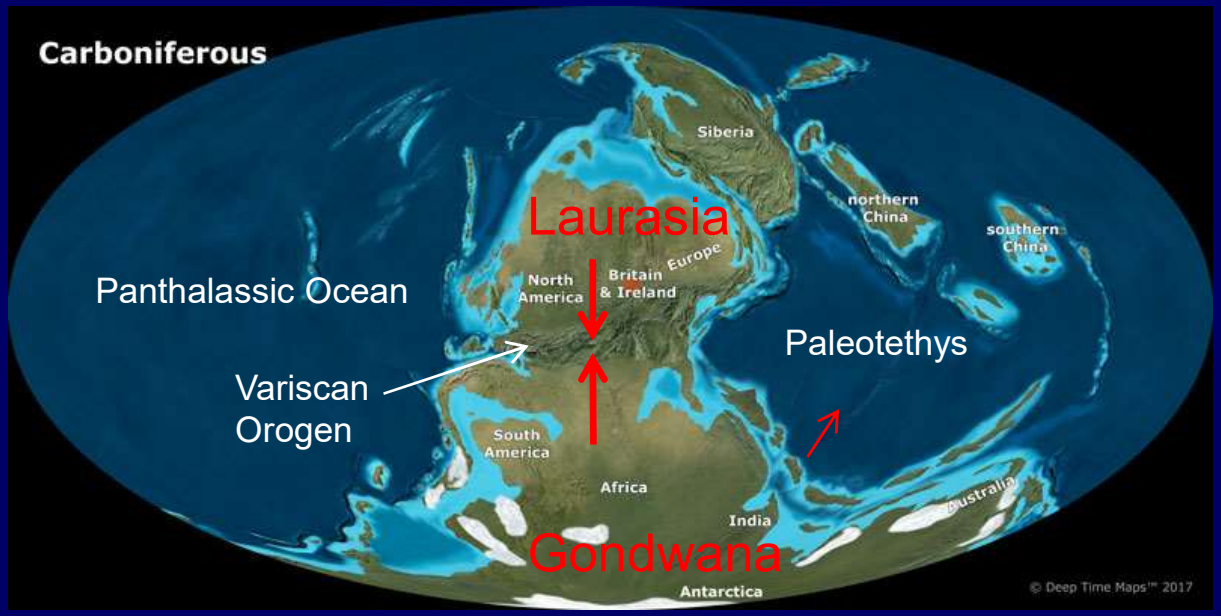
Cretaceous chalk

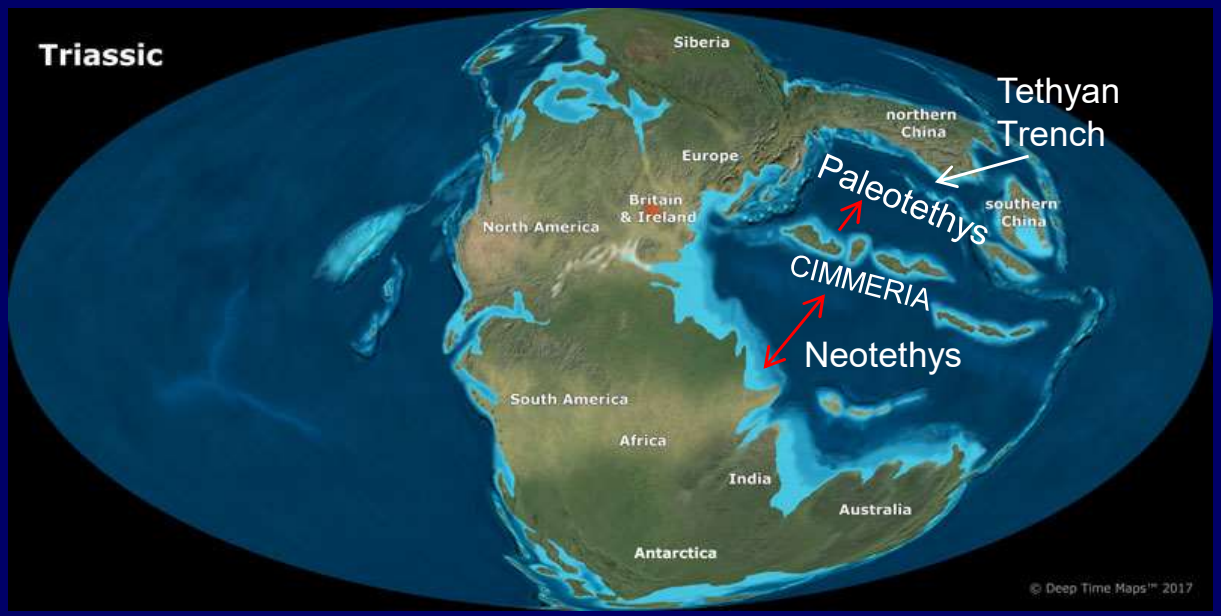


Limestones of the Jura Mountains, Switzerland

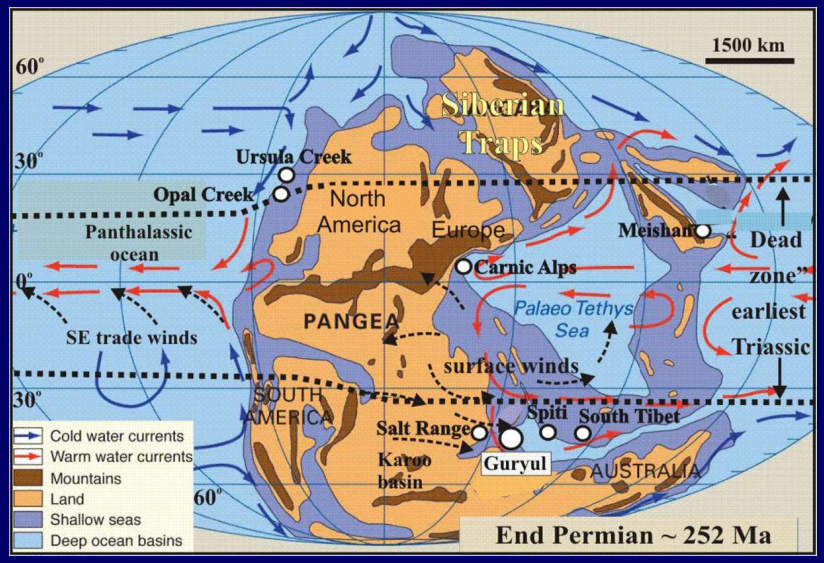
Triassic = 50.5 myr duration
Jurassic = 56.4 myr duration
Cretaceous = 79 myr duration

Pangaea and Triassic (251.9 – 201.4 Ma) palaeogeography





The collision of the two supercontinents, Laurussia and Gondwana in the late Carboniferous (Variscan Orogeny) created the single supercontinent Pangaea



Supercontinent size and location influenced global climate and ocean circulation patterns

Palaeogeographic reconstructions, showing the supercontinent Pangaea

Causes of Phanerozoic mass extinctions





The beginning and end of the Triassic are marked by major extinction events (P-T and T-J)

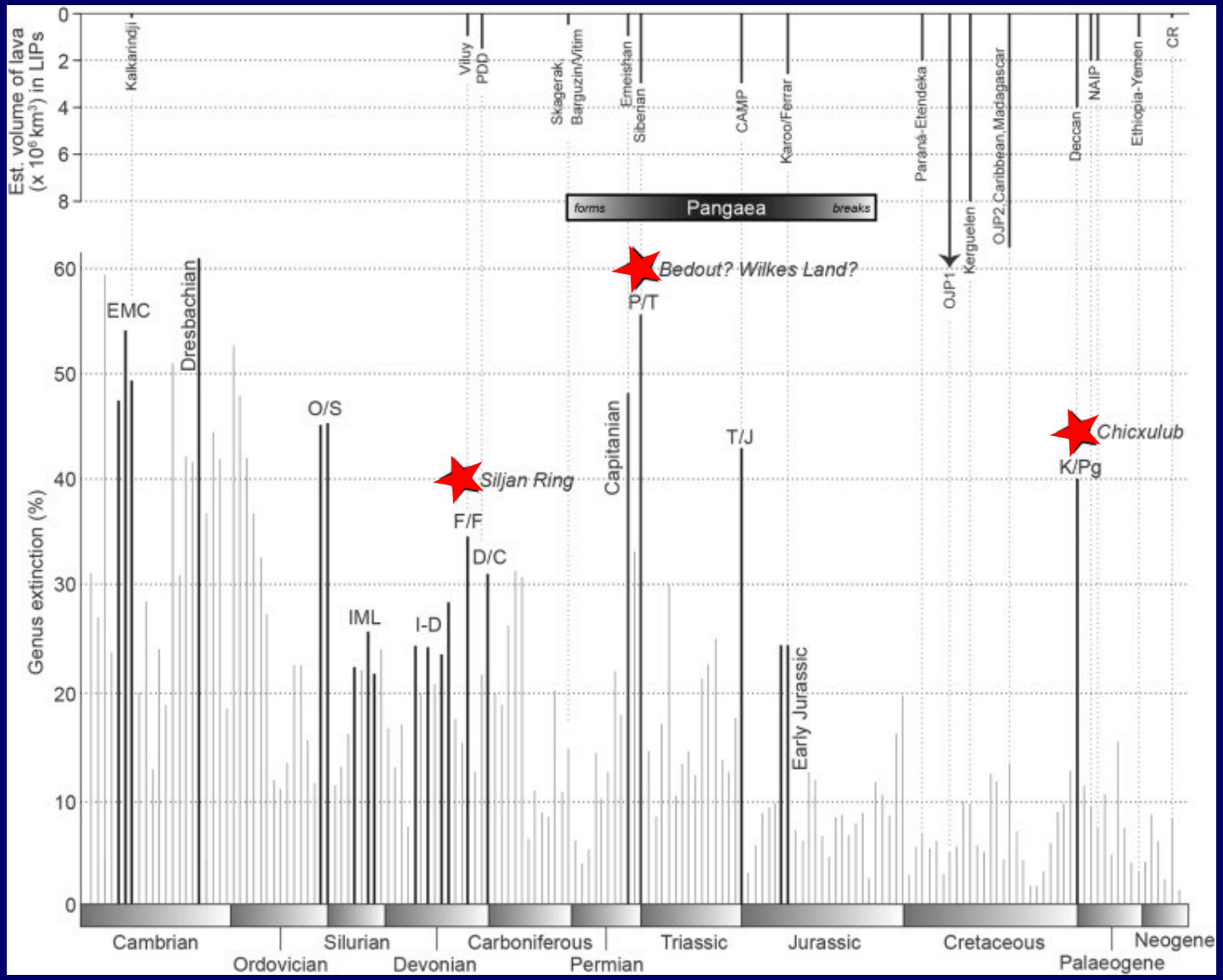
The causes are a matter of debate: meteorite impacts vs massive volcanic eruptions

Causes of Phanerozoic mass extinctions

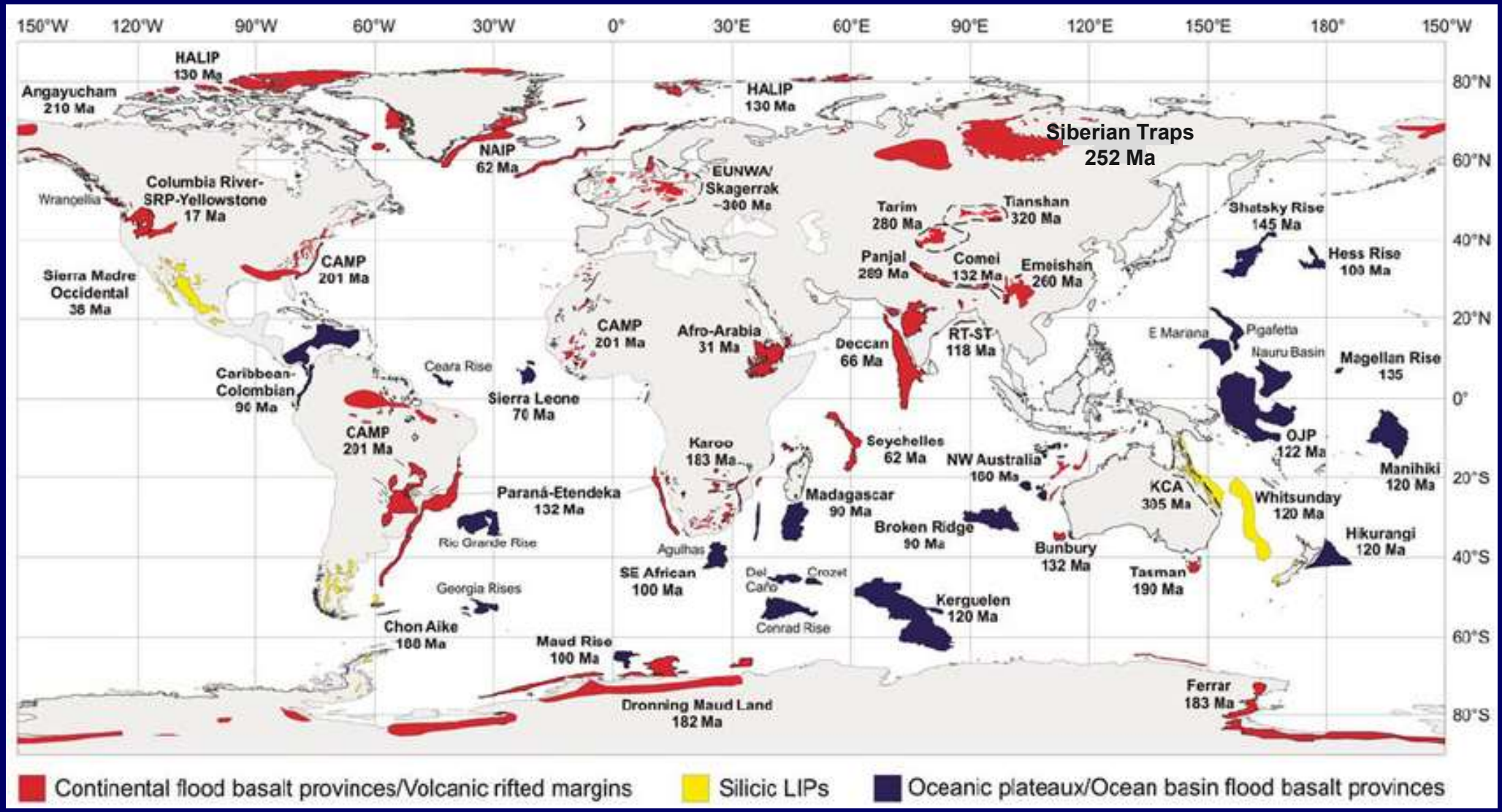
One view point: Hyperthermals, not asteroid impacts, is the cause of most mass extinctions through the Phanerozoic

Hyperthermals: rapid and extreme global warming events in our geological past driven by massive volcanic eruptions

Genus extinction magnitude through the Phanerozoic versus the age and estimated original volume of LIPs, and the age of supposed bolide impact structures (★)

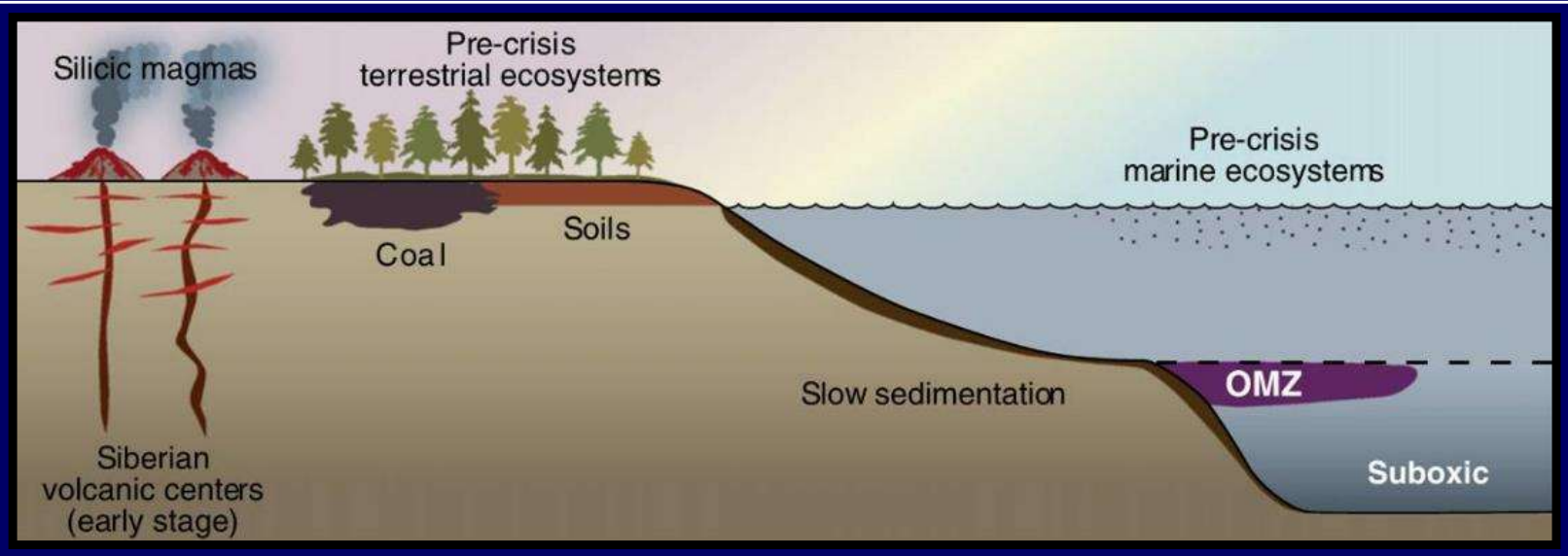


Global distribution of Mesozoic LIP volcanism



Large Igneous Provence (LIP) volcanism is believed to be associated with mantle plumes

Permian-Triassic extinction: volcanic causes



Just before the extinction

OMZ = oxygen minimum zone (200-1000 m depth)

Eruption of 252 Ma Siberian Traps

- 3 million km^3 of lava erupted in bursts over 600 kyr

Believed to be ultimate cause of PTB mass extinction

- severest crisis in Phanerozoic, affecting >90% marine species, 70% land vertebrate genera and most land vegetation

But, eruption of the flood basalts between 251.7 - 251.3 Ma: significantly *after* the main PTB extinction horizon (252.28 ± 0.08 Ma)

- Intrusion of sill complex trigger for mass extinction – age of sills 251.9 – 252.2 Ma predate lavas
  - sill intrusion in volatile-rich sediments liberated massive greenhouse gas volumes
- Ignimbrite flare-up – intense silicic volcanism characterized by multiple caldera-forming eruptions that together evacuate 10^3 to 10^4 km^3 of magma. Associated with subduction volcanism during Early Triassic closure of the Paleo-Tethys Ocean

How do Large Igneous Province (LIP) eruptions cause mass extinctions?

Model for LIP*-induced mass extinctions

*LIP- Large Igneous Province

'Standard model' for mass extinctions throughout Phanerozoic (through Permian, Triassic, and Jurassic)

Injection

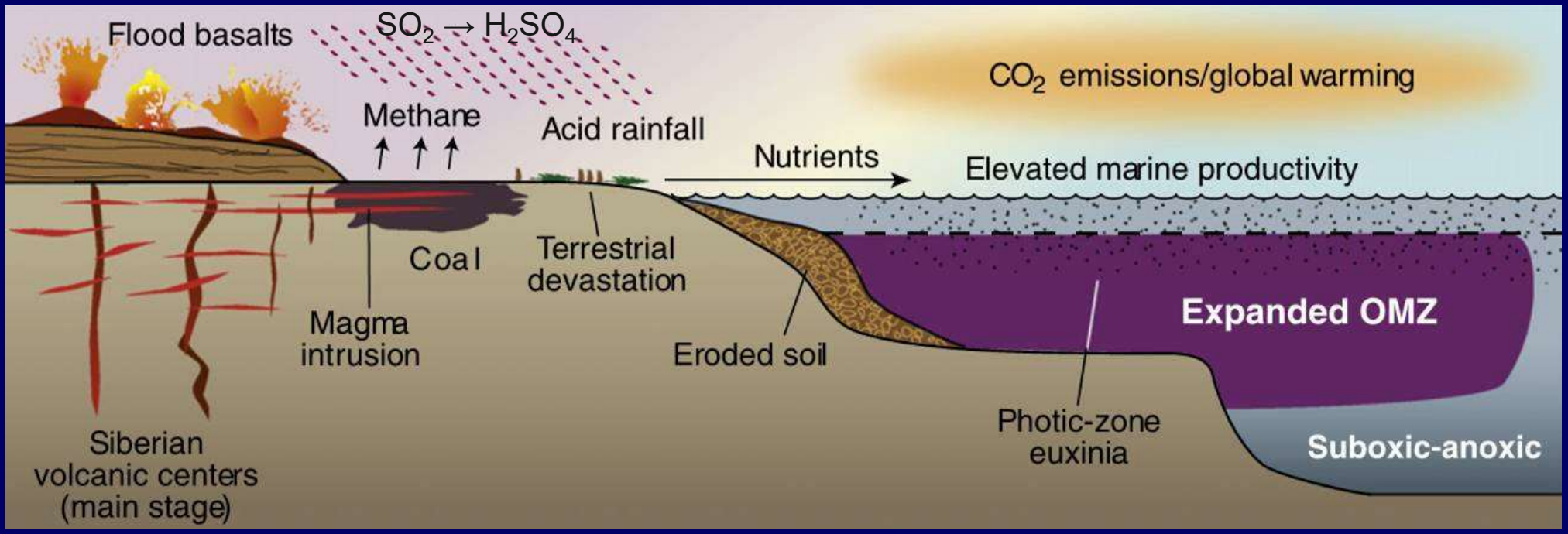
CO_2 H_2O

SO_2 HF

Ash

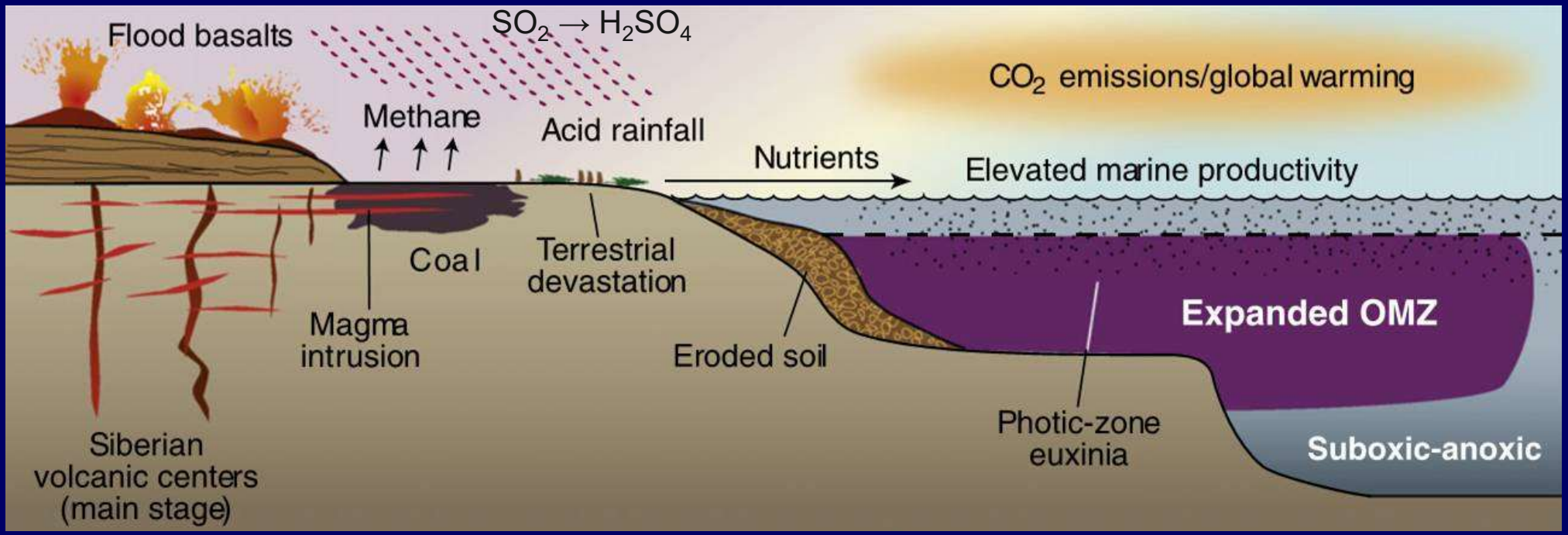
HCl

Key elements of environmental impact: acid rain, rapid and extreme global warming, ocean anoxia + secondary contributions from ozone depletion, toxic metal poisoning



Increased atmospheric greenhouse gas levels (direct CO_2 emissions + methane emissions from m/m coal, organic-rich mudstones and evaporites) → global warming

Model for LIP-induced mass extinctions





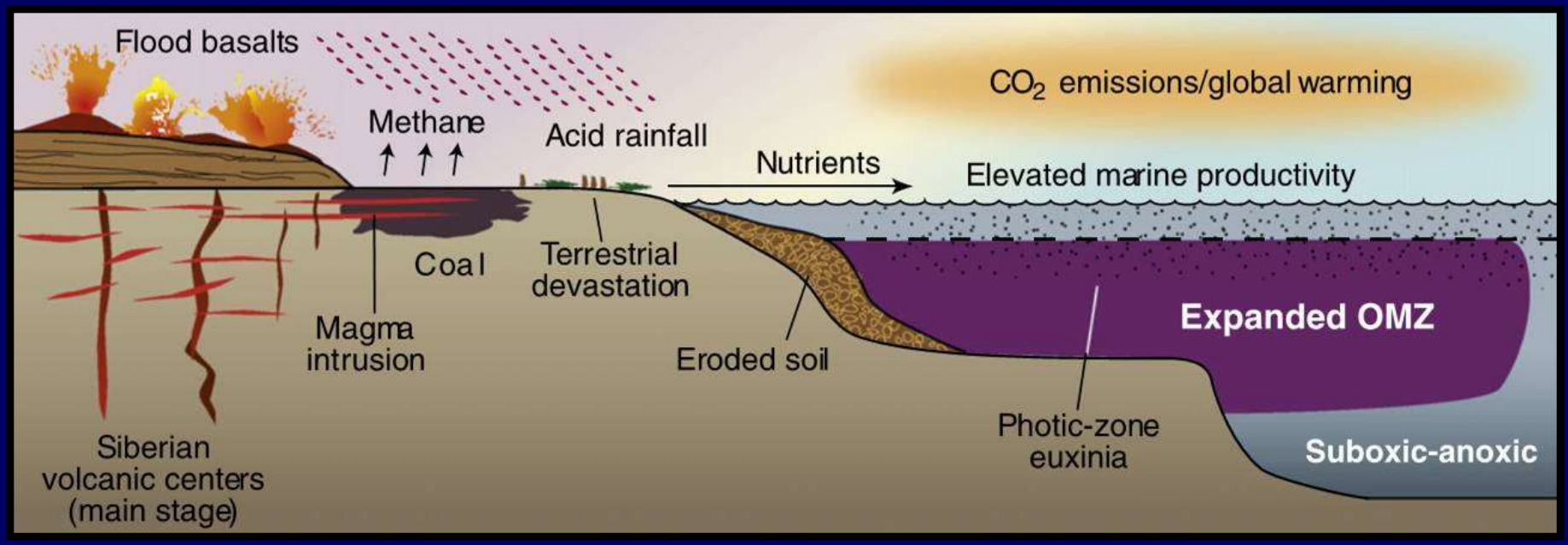
SO_2 emissions → mixing with H_2O in atmosphere → acid rain (sulfuric acid)

Rapid and extreme warming & acid rain → devastation of vegetation (pH of soil, effect on photosynthesis) and drought → reduction in habitable regions, disruption of the food chain & effects of extreme heat → terrestrial mass extinction

Model for LIP-induced mass extinctions

Reduced vegetation cover & acid rain → increased continental weathering (& stripping of soil) → (7-fold) increase in sediment input to ocean → benthos overwhelmed

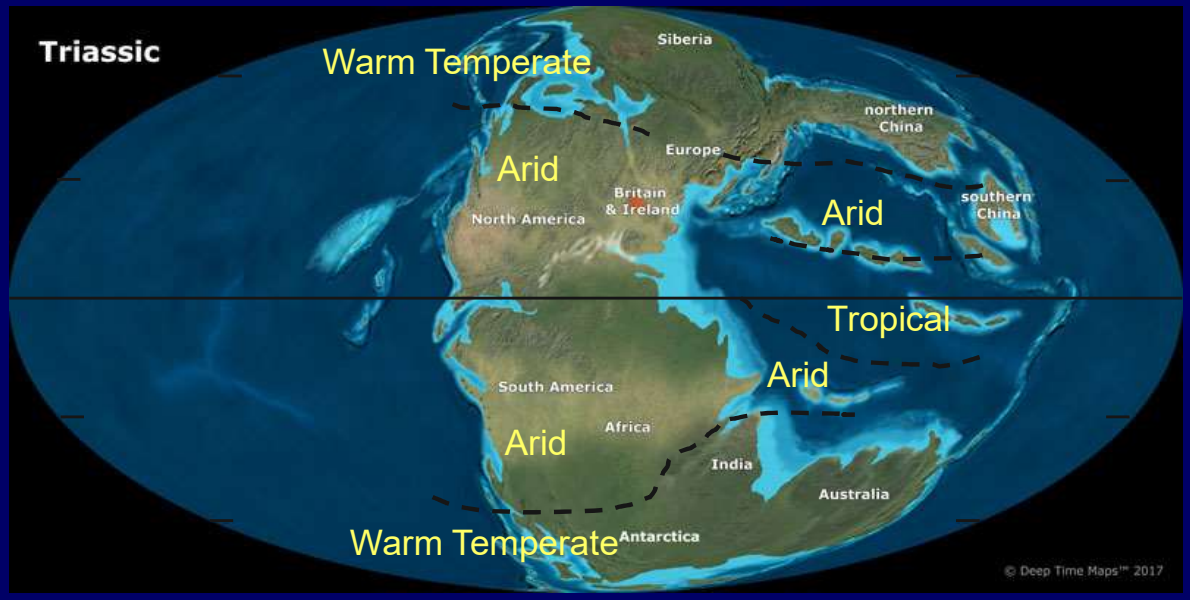
Increased sediment to ocean → increased nutrient input → enhanced marine surface productivity (with sluggish oceanic circulation) & increased temperature (e.g., less dissolved oxygen) → reduced oxygenation of oceans & increased acidity (upwelling of CO_2 from anoxic deep water) → narrow to absent habitable depth zone → marine mass extinction

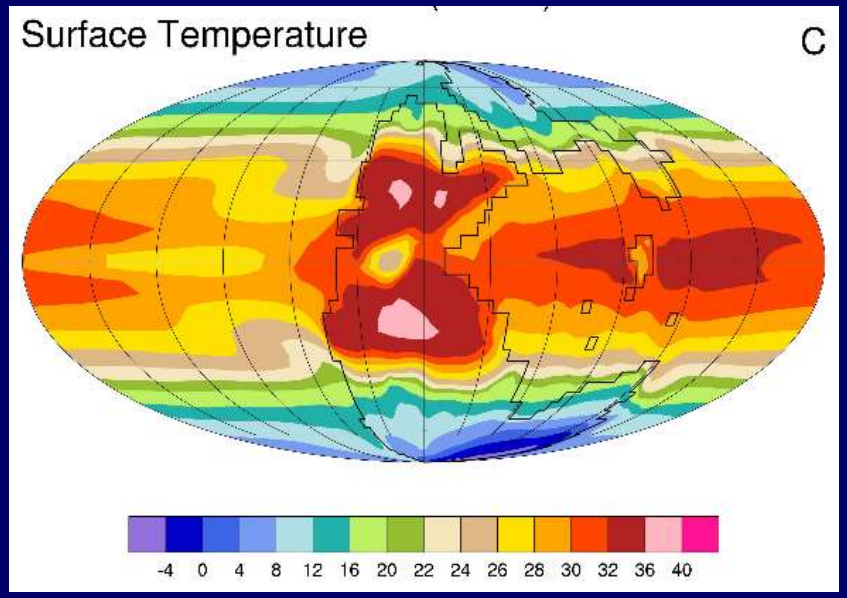


Become 'Standard model' for mass extinctions throughout Phanerozoic (through Permian, Triassic, and Jurassic)

Early Triassic aftermath of main stage Siberian eruptions

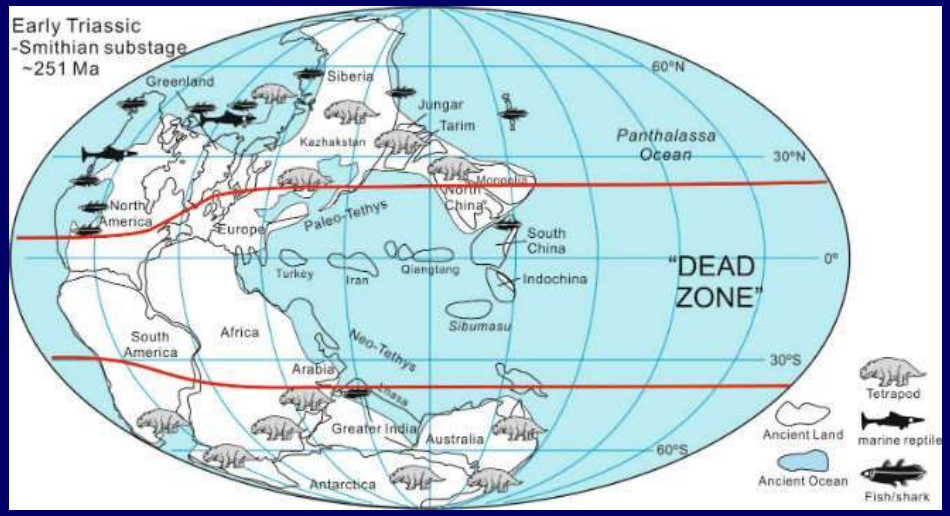
Very poor environmental conditions for 6 myr into the Triassic with several episodes of intense global warming (ongoing pulses of Siberian Trap eruptions)

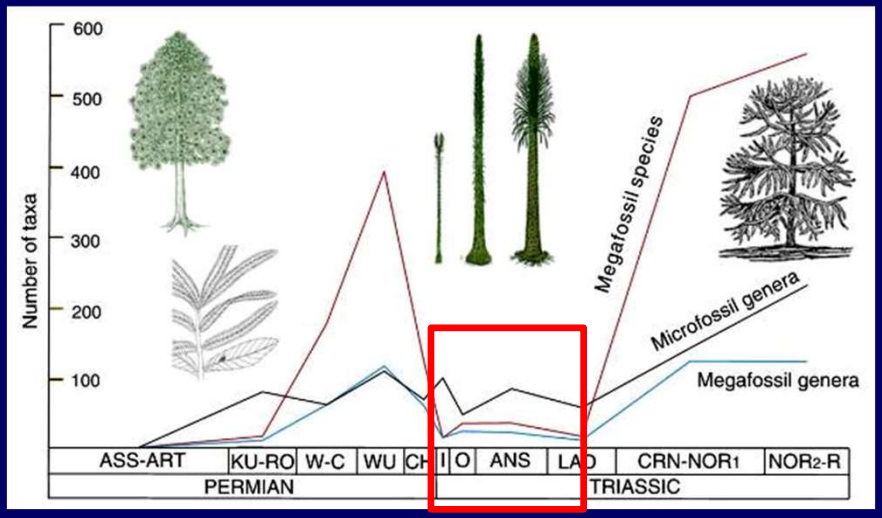




- major expansion of the semi-arid zones to latitudes of 70˚
- large increase in temperatures in polar regions (no polar ice caps)
- rapid rise in surface temperatures to lethal levels in the tropics: some models predict ~ 50˚- 60˚C on land, and 40˚C at the sea-surface (modern equatorial sea surface temperatures are ~28-30˚C and are never above 32˚C)

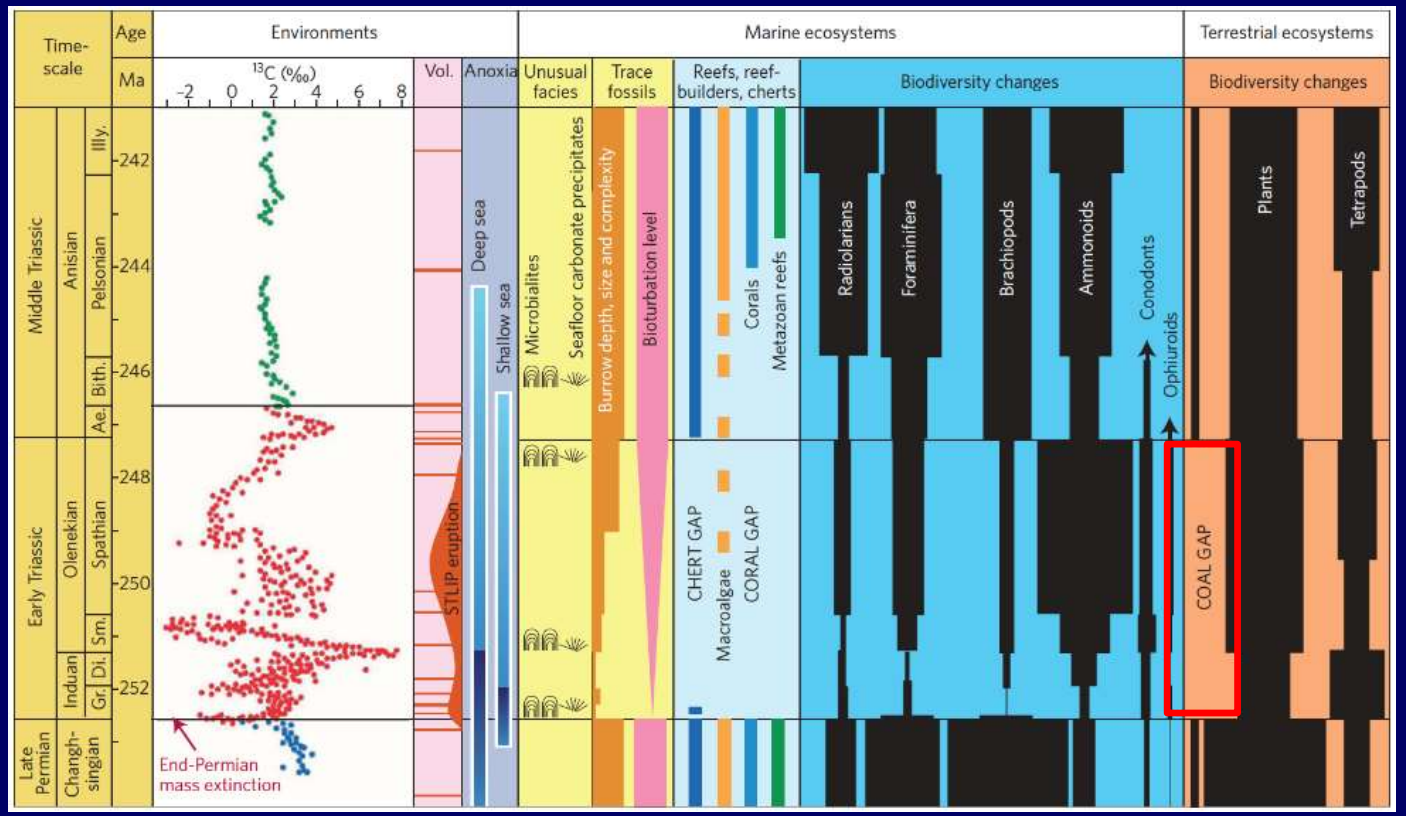
Early Triassic aftermath of end Permian extinction





Loss of 90% of species globally including loss of vertebrates (fish, marine reptiles, tetrapods) in equatorial zones

~10 myr "coal gap" suggesting the termination of peat-forming floras



Early Triassic aftermath of end Permian extinction

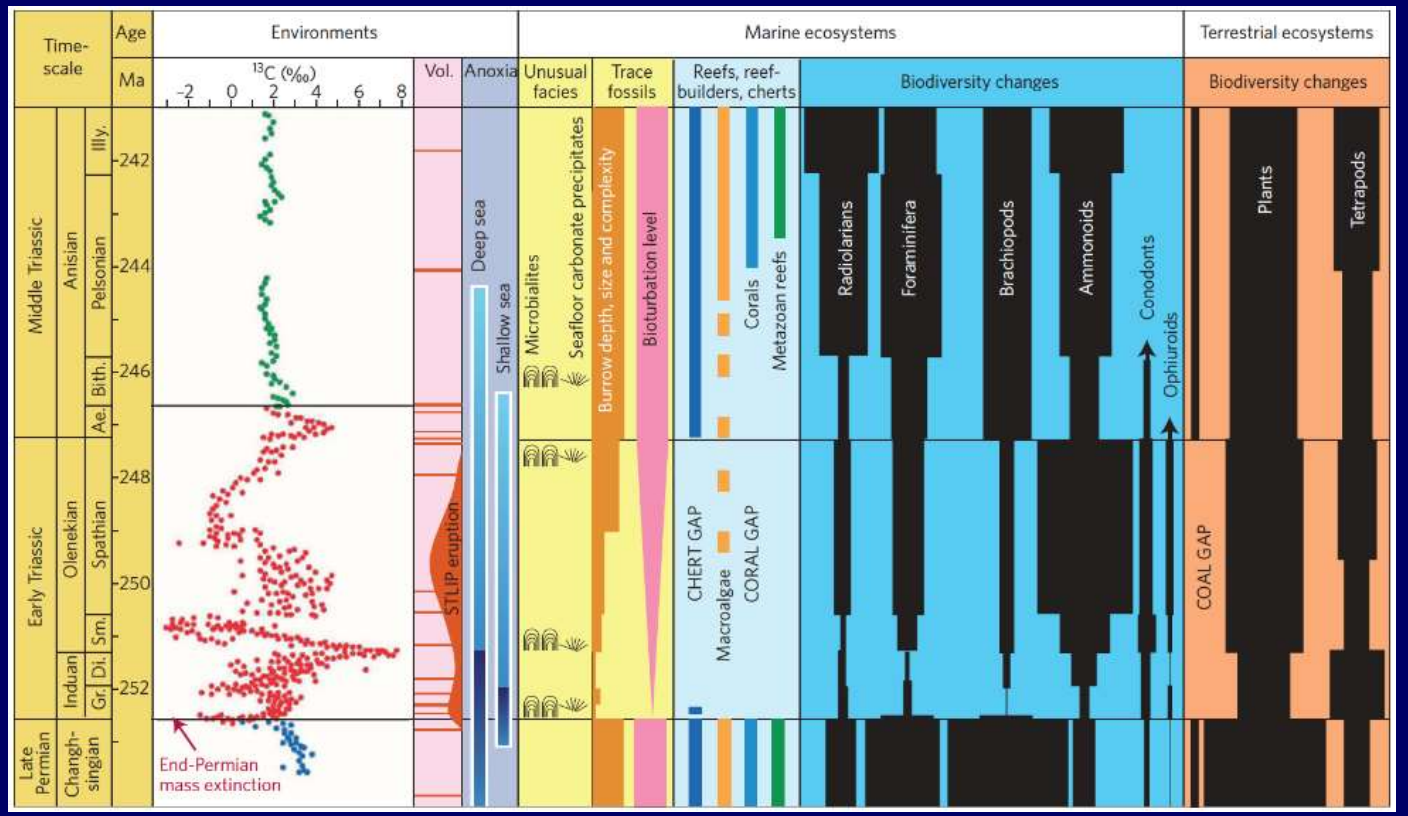
Marine ecosystem far from normal:

- absence of metazoan reef builders (e.g., corals)
- absence of deep-sea cherts

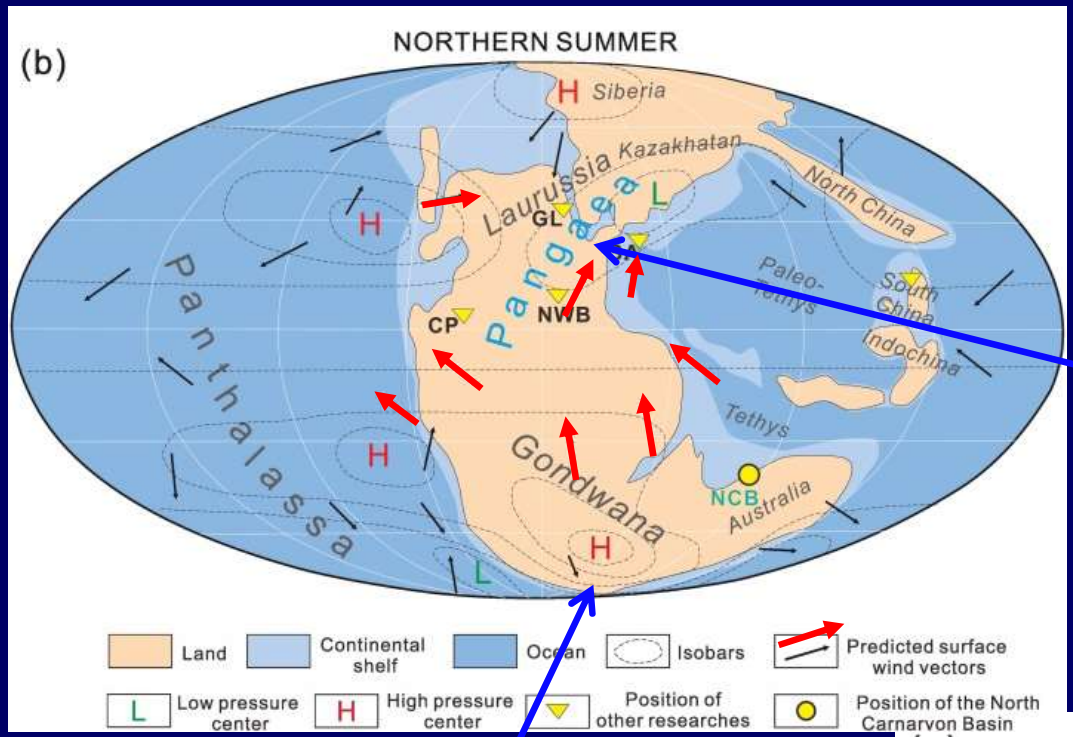
Ongoing greenhouse condition:
Enhanced weathering → increased sediment to ocean → increased nutrient input → enhanced primary productivity & eutrophic conditions

→ favour stromatolitic microbes not reliant on oxygen over corals and filter feeders

Biotic diversity did not begin to recover to pre-extinction levels until the Middle Triassic



Effects of Pangaea on the Triassic-Jurassic climate



The distribution of the land masses & equatorial Tethys ocean led to the development of a mega-monsoonal climate

- distinct wet and dry seasons associated with seasonal reversal of winds

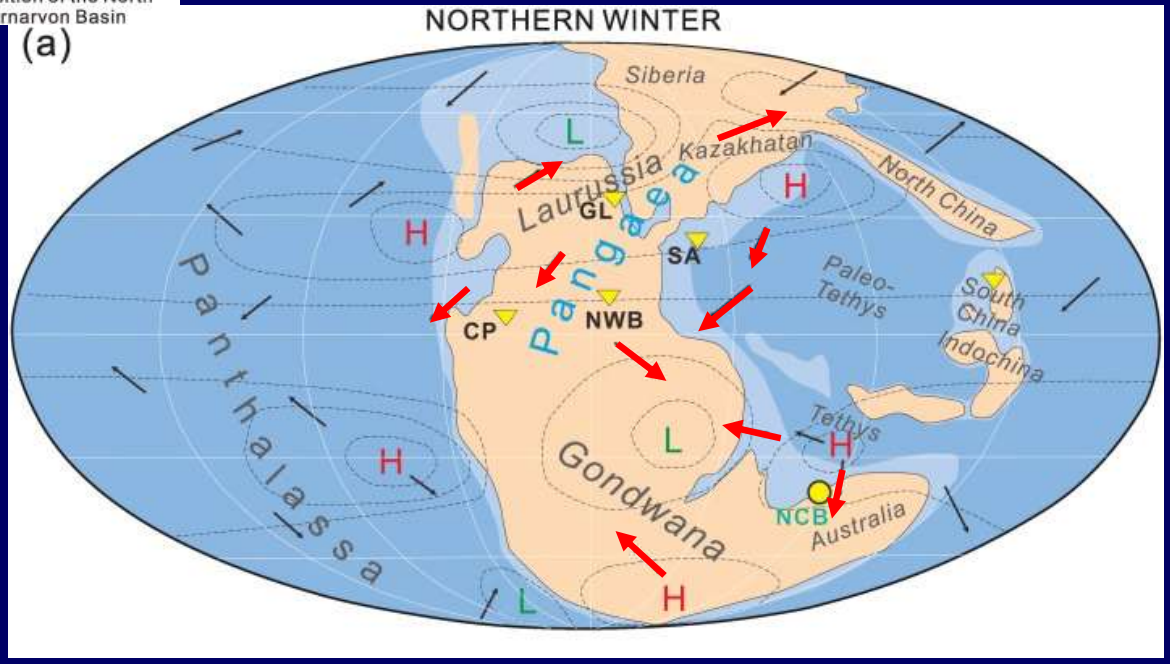
Atmospheric flow from H → L pressure

- air flow over the warm Tethys Ocean where it would have picked up moisture

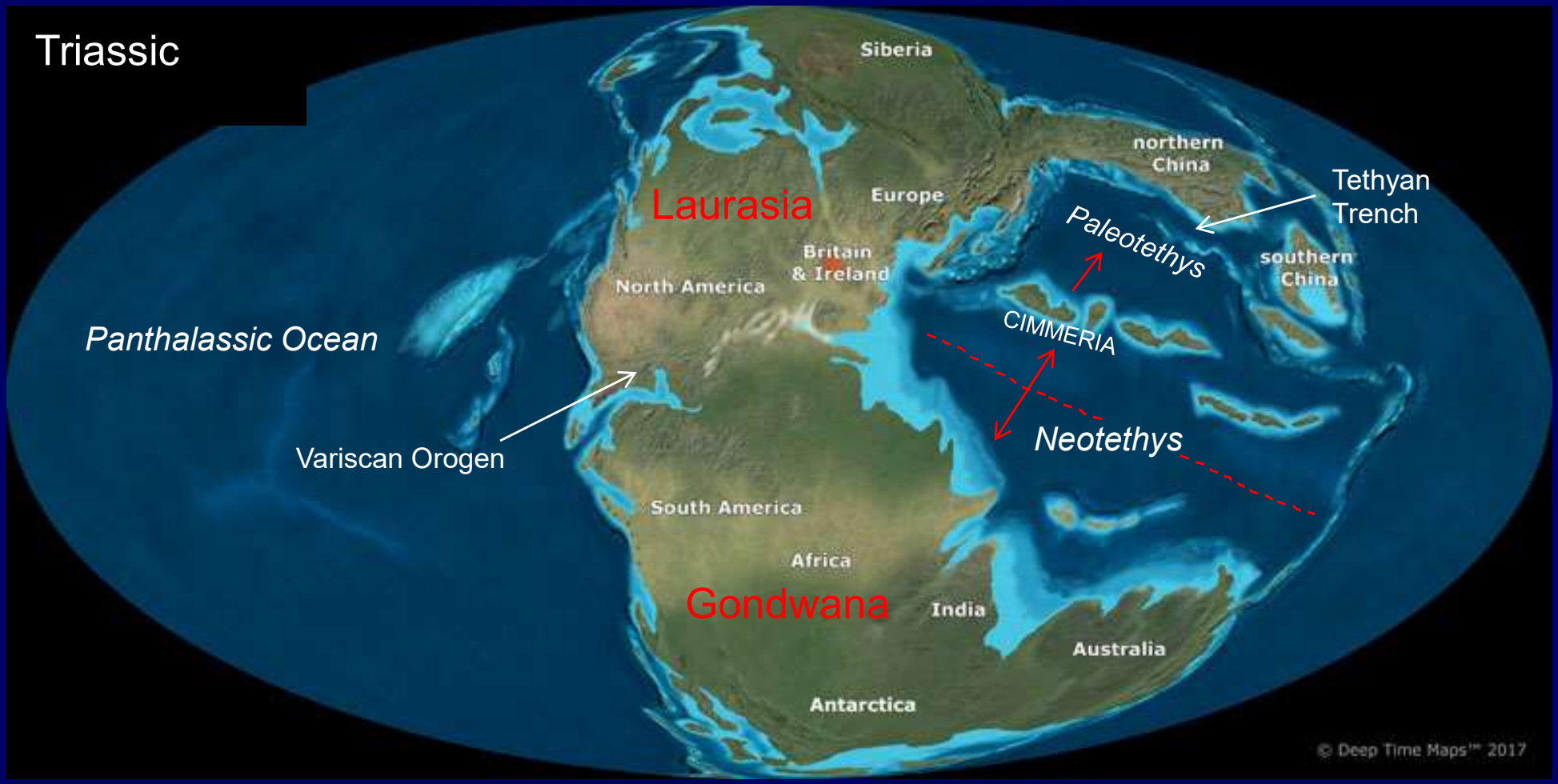
→ area of intense precipitation over Laurussia

Vegetated polar regions → ice limited & seasonal

Pattern of winds and rainfall reversed in winter



Monsoonal conditions in the late Triassic



Monsoonal conditions probably initiated in the Permian once Pangaea was formed and had drifted over the equator

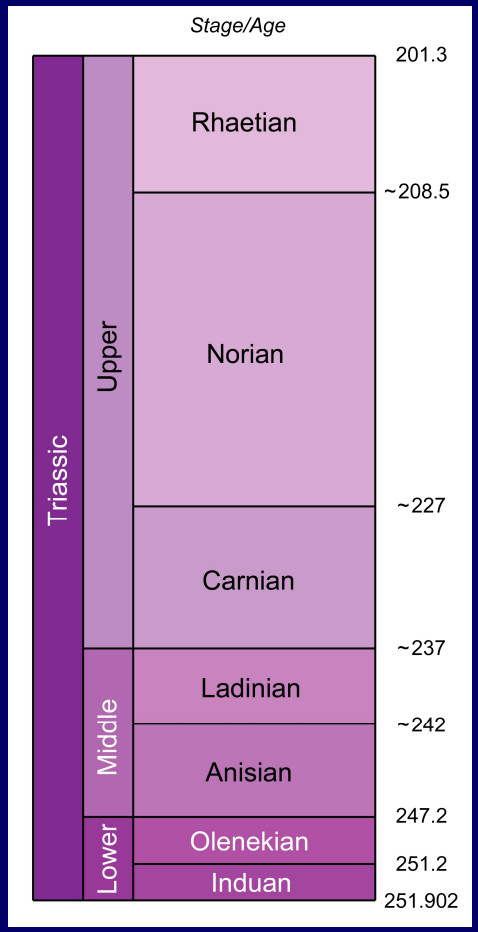
Conditions intensified from the late Triassic once the Cimmerian terranes had collided with Laurasia and Neotethys had become free of land masses

As Pangaea drifted north and began to break up during the Jurassic, the mega monsoonal conditions weakened

Triassic pluvial intervals: the Triassic was not just hot and dry!

Pluvial event: periods of enhanced wetness

Greenhouse crises





- Carnian Pluvial Episode: intensely humid interval
- eruption of Wrangellian flood basalts (~ 230 Ma)

→ double pulse of climate warming (~ 7˚C)

→ v. high rainfall → increased siliciclastic influx to the oceans → overwhelmed shallow marine reef environments → the shutdown of marine carbonate systems worldwide

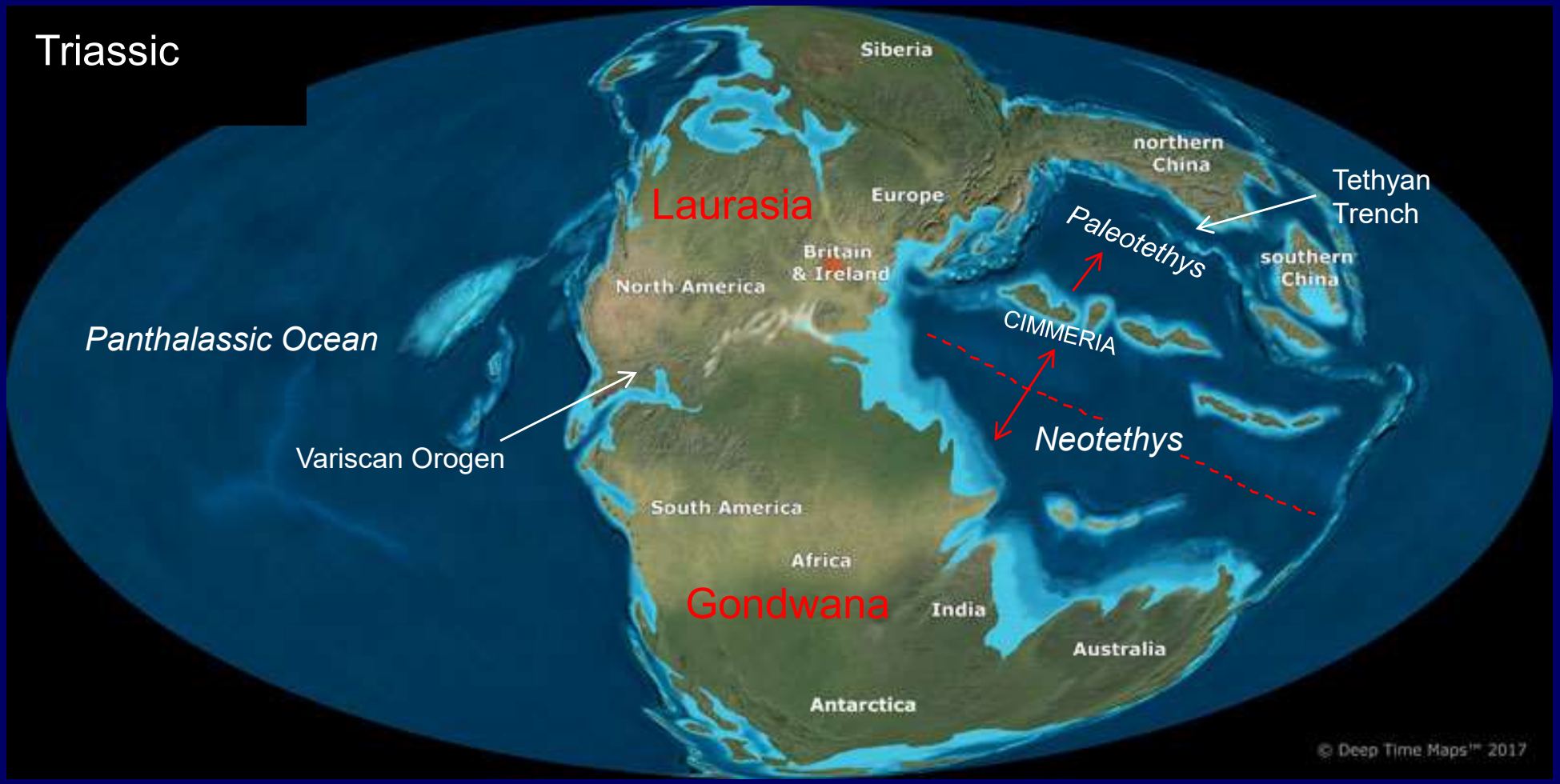
→ intense and widespread oceanic anoxia

→ extinctions in ammonoids, crinoids, bryozoa, conodonts

Good news!!! Major extinctions are followed by major evolutionary innovations

e.g., Carnian extinctions followed by rise of dinosaurs, first flying vertebrate, spread of conifer-dominated floras worldwide, appearance of calcareous nannoplankton which now dominate the global carbon cycle, early mammals

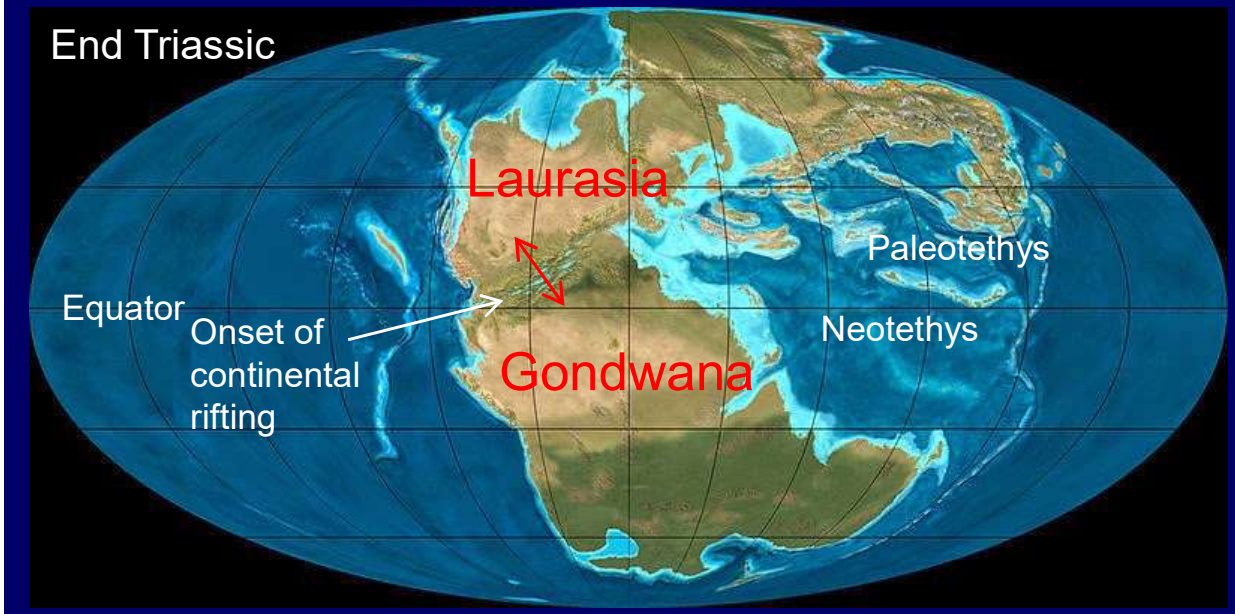
After 100 myr comes the break up of Pangaea



Break up of Pangaea occurred in three main phases

1. beginning Early Jurassic rifting between Laurasia and Gondwana
2. beginning Early Cretaceous: Break-up of Gondwana into Africa, Sth America, India, Antarctica & Australia
3. Early Cenozoic rifting of Laurasia (aka Laurentia = Nth America & Greenland) rifted from Eurasia

Phase 1: Onset of continental rifting of Pangaea

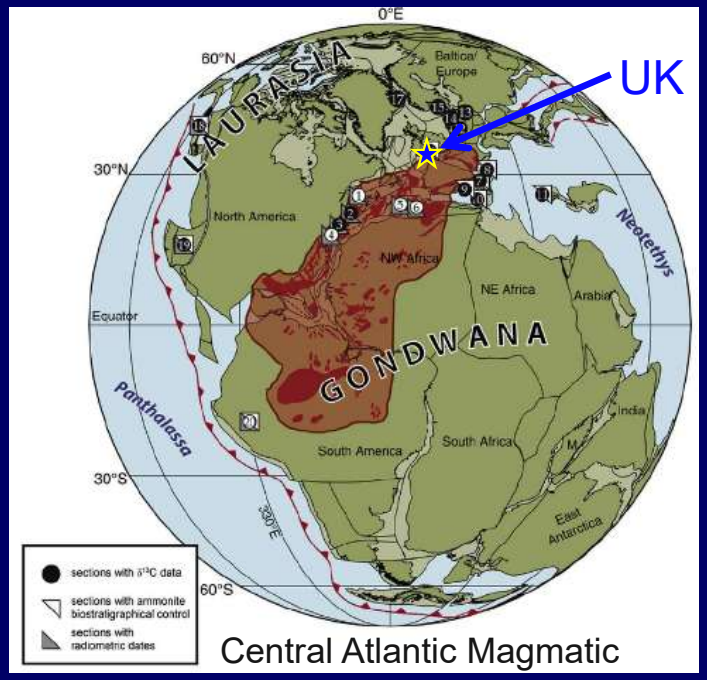


Rifting at center between Laurasia and Gondwana produced:

- multiple failed rifts
- incipient Central Atlantic Ocean
- emplacement of subaerial basaltic lavas and intrusives sills and dykes of CAMP

Central Atlantic Magmatic Province (CAMP)

- most aerially extensive and voluminous (2-3 million km^3) Large Igneous Province on Earth
- erupted in 4 bursts 202-220 Ma



Central Atlantic Magmatic Province



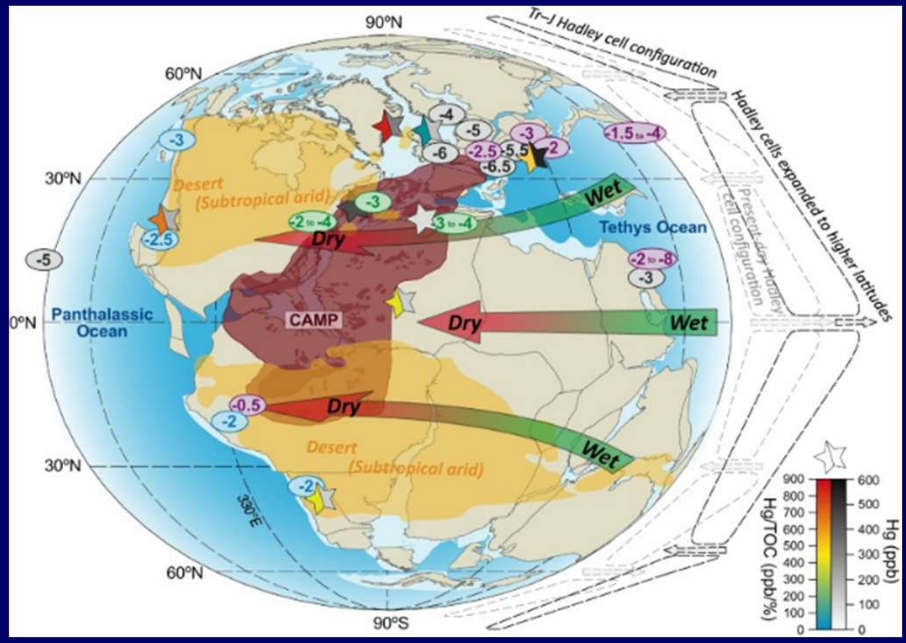
CAMP flood basalt (Agouim High Atlas, Morocco)

CAMP & the Triassic - Jurassic extinction event

The end of the Triassic saw one of the big five mass extinctions, affecting both marine and terrestrial biota

Synchronous with the first pulse of volcanic eruptions in the Central Atlantic Magmatic Province

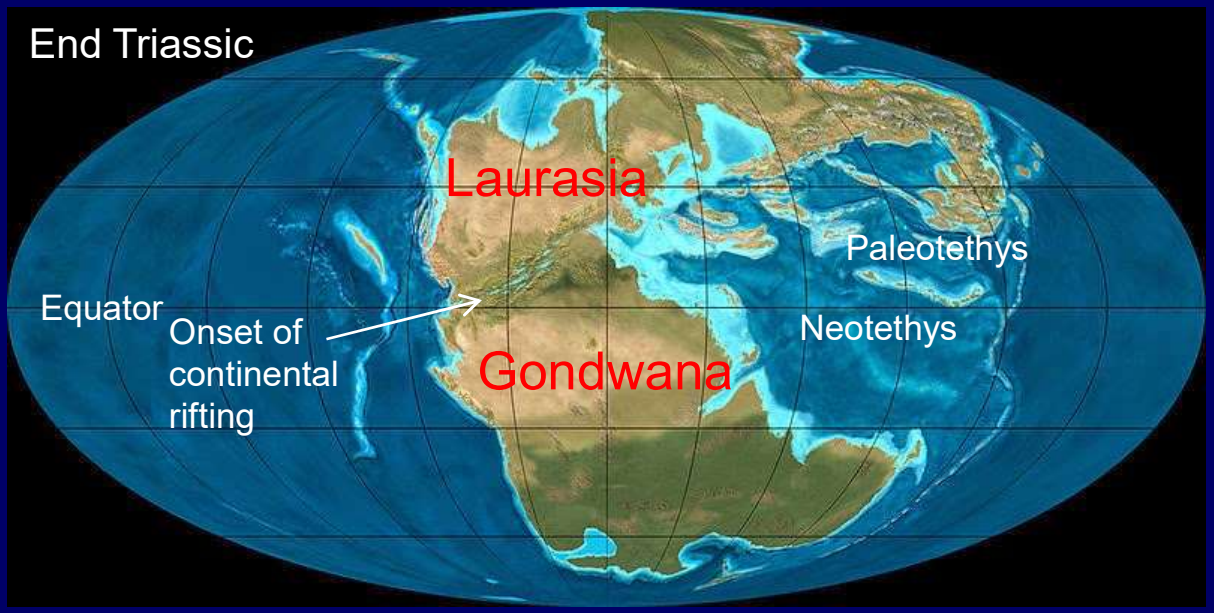
Significant ocean acidification → selective loss of shelly marine forms (extinction of reef communities, severe effect on ammonites, bivalves, calcareous microfossils, conodonts)



Climate and distribution of CAMP volcanism at the end of the Triassic

Four-fold increase on atmospheric CO_2 and global warming ~5˚C → dramatic turnover in pollen, spores and vertebrates

Opening of the Central-Atlantic seaway



By the Mid-Jurassic oceanic crust began to form

Once rifting resulted in the formation of a new ocean, a seaway between the Tethys Panthalassic Ocean was formed

Central Atlantic sea way



Jurassic extension and Cretaceous sedimentation



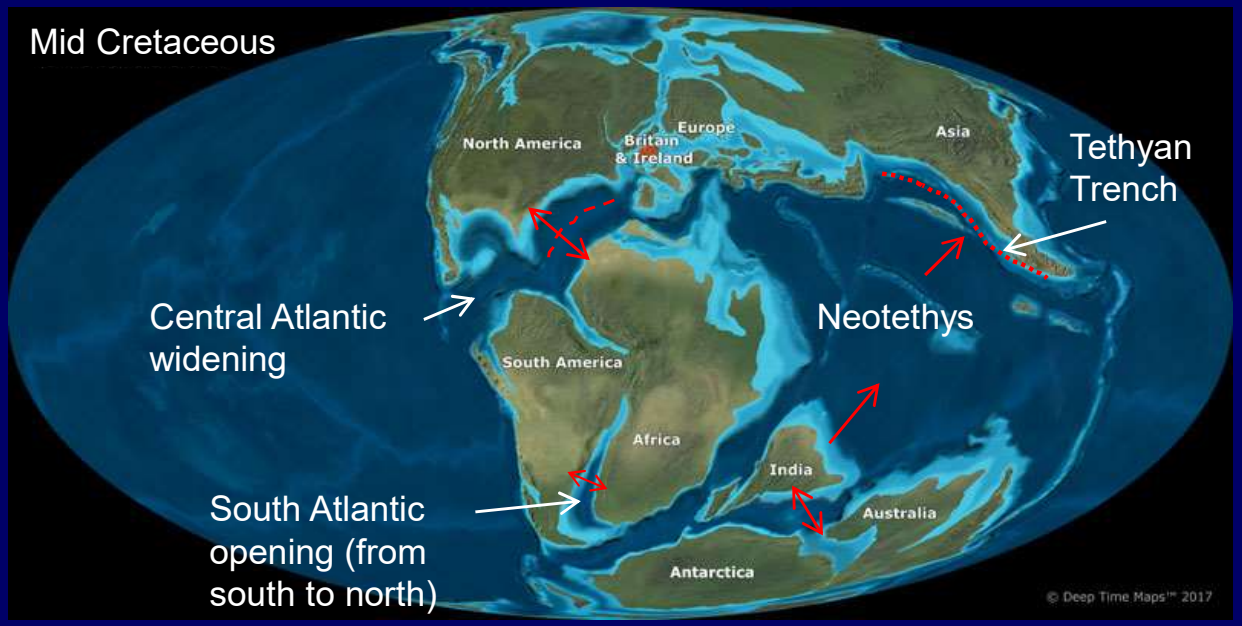
Jurassic extension initiated the formation of the basins into which Cretaceous sedimentary units were later deposited

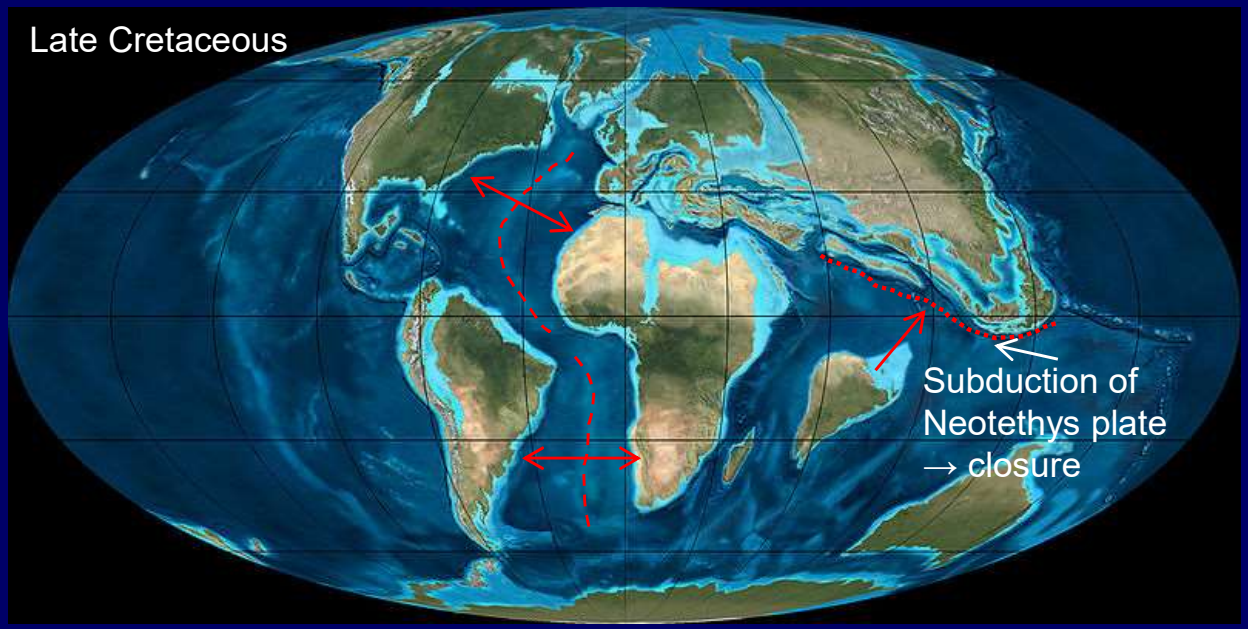
e.g. into the area that is now the Mediterranean



Barreme field course, France

Cretaceous plate configuration





During the Cretaceous, the Central Atlantic continued to widen

Early Cretaceous Gondwana began to break apart

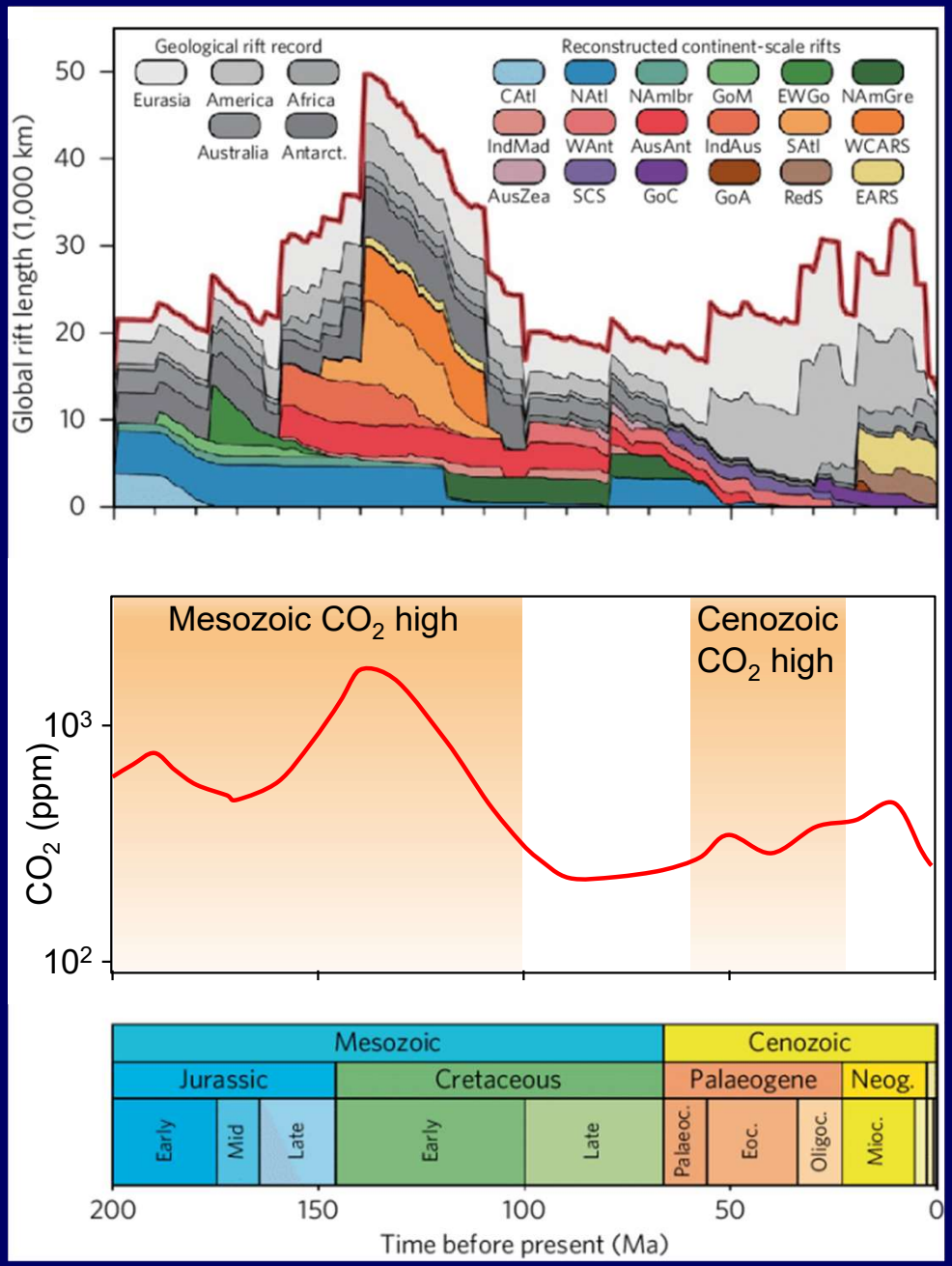
- South America and Africa separated from southern Gondwana
- India began to separate from Antarctica and Australia

By the mid-Cretaceous the South Atlantic was starting to open from south to north

Late Cretaceous subduction of Tethyan Trench

- India, Africa and Australia drift north
- subsequent rapid passage of India and Africa towards Eurasia was to lead to closure of Neotethys

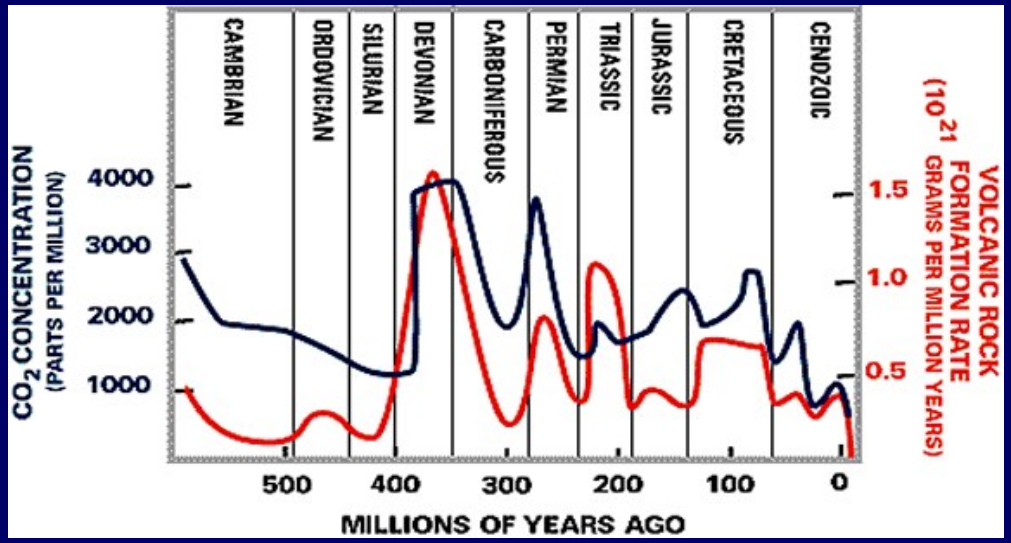
Effect of Pangaea breakup on climate



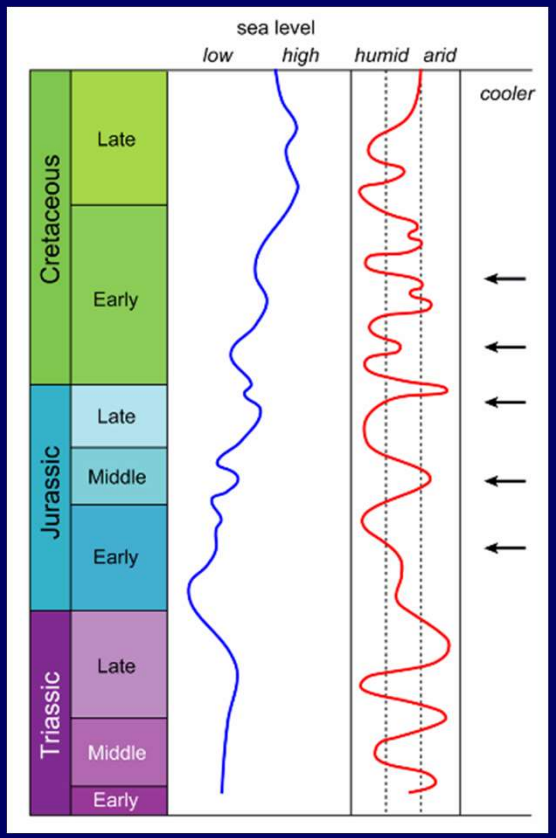
Pangaea breakup → active sea floor spreading → increased volcanism → CO_2 and SO_2 & faster recycling of subducted carbonates → accelerated production of CO_2
→ greenhouse climate

Atmospheric CO_2 ~ 4x present levels during Jurassic, ~ 10x present levels during Cretaceous → global temperatures much warmer, more humid

Land mass at low to mid-latitudes → small pole-equator thermal gradients (no polar ice)



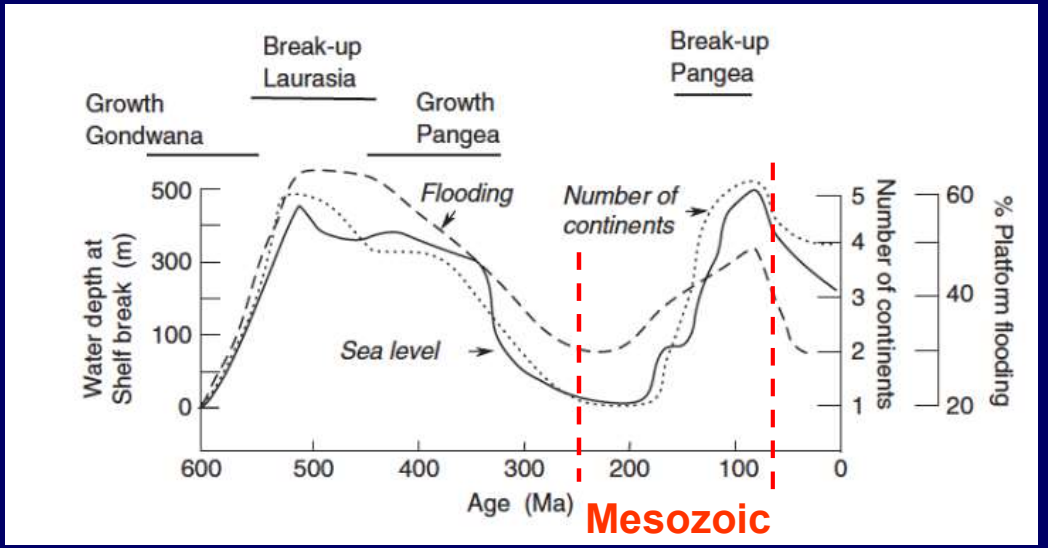
Effect of Pangaea breakup on sea level



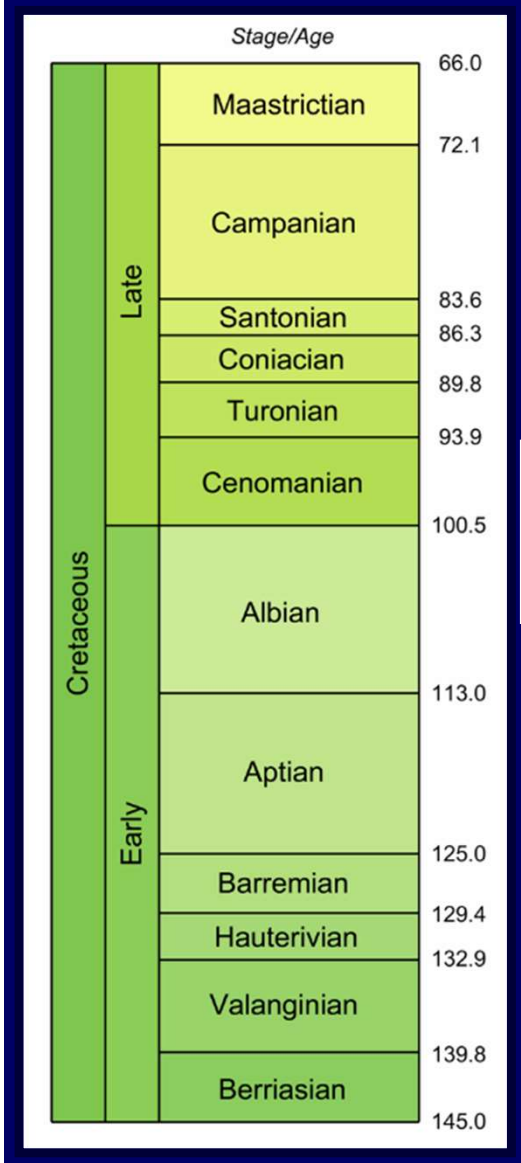




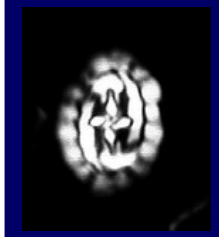
Gradually rising global sea levels (with fluctuations) through the Jurassic and Cretaceous to become the highest in the Phanerozoic (>200 m above today) → flooding of lowlands (transgression) to form vast, warm inland seas (exposed land 60-70% of present day)



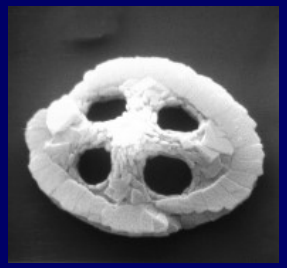
The rise of calcareous nannoplankton (Coccolithophorids)

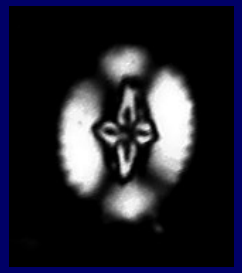


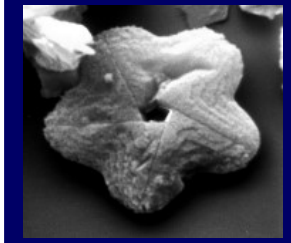
Biostratigraphic range

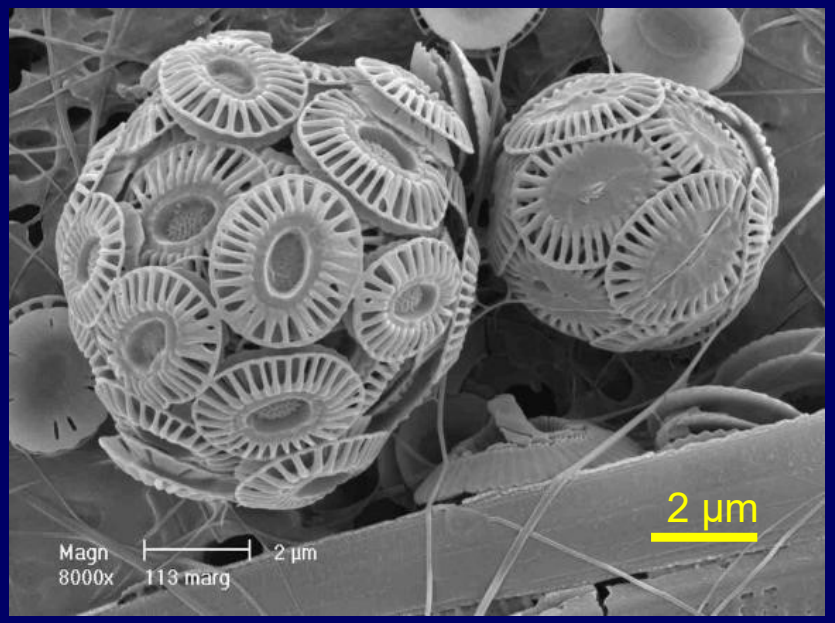












SEM photograph of *Emiliyana huxleyi* (a modern coccolithophore). Note the overlapping plates (coccoliths) each about 0.002 to 0.004mm across

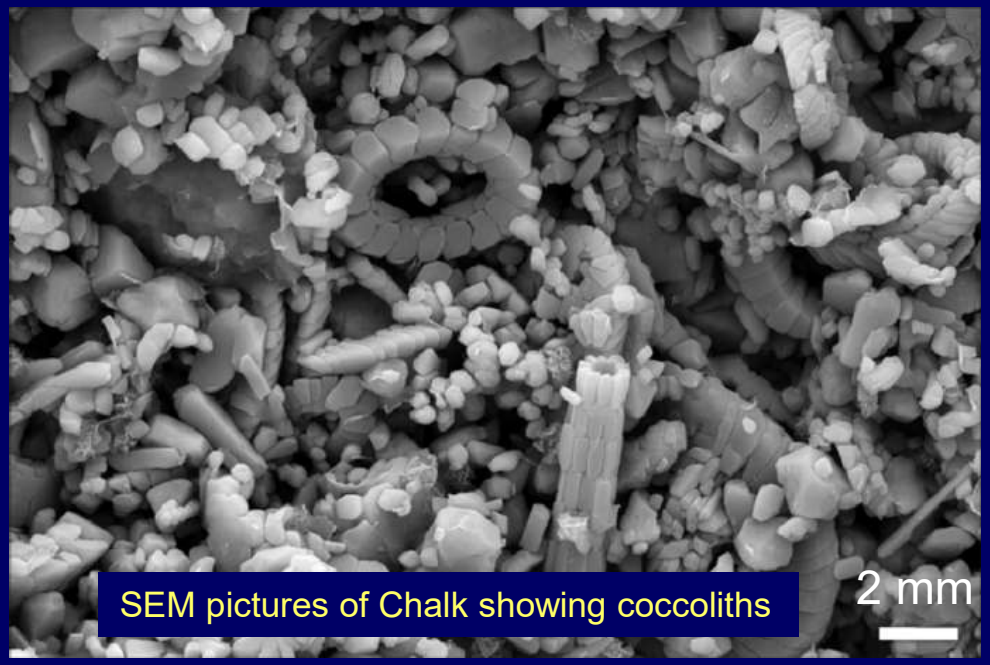
Coccolithophores

- first appeared after Carnian Pluvial Episode (late Triassic) in epicontinental seas
- consist of an algal cell surrounded by calcite plates called coccoliths

Over half carbonate removed from ocean today is extracted by coccolithophorids

Abundant floating in near surface water of the oceans (planktonic inhabiting the pelagic zone)

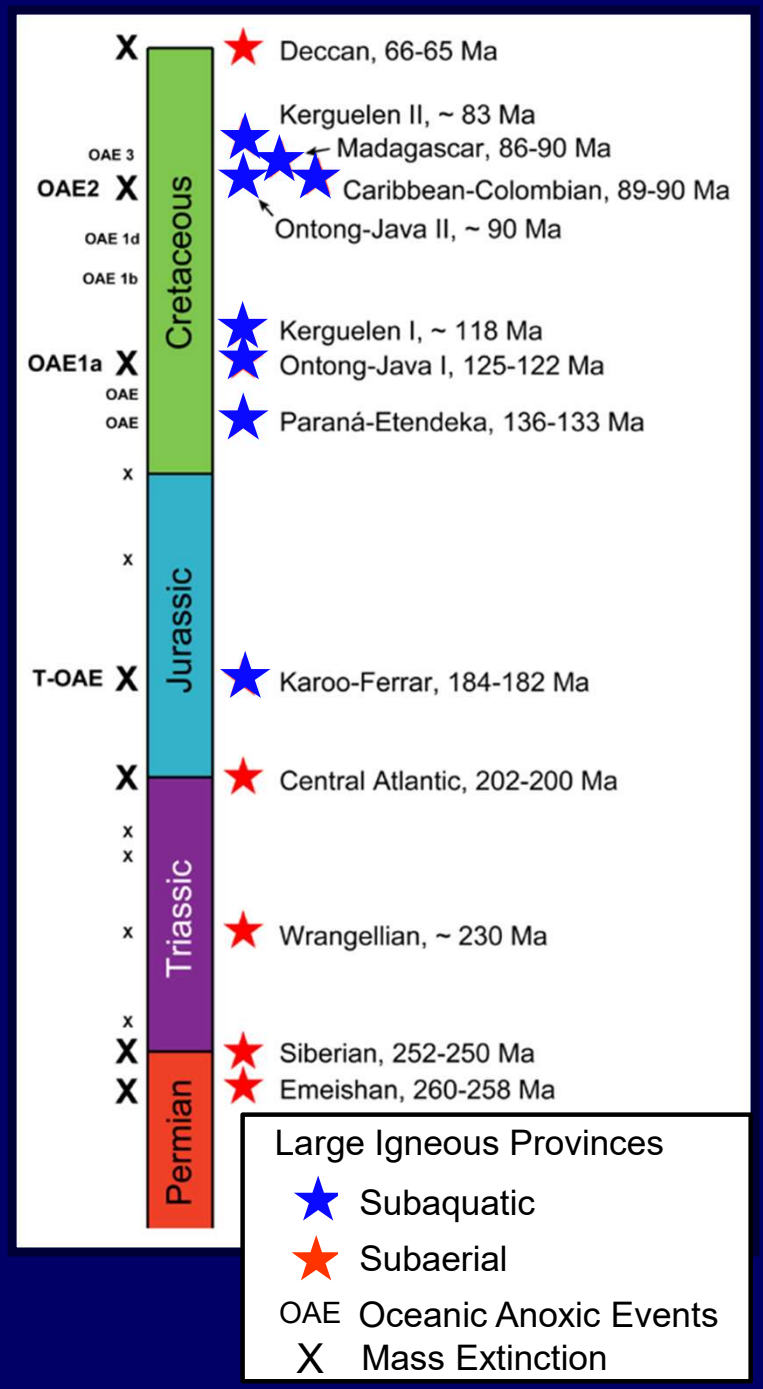
Deposition of Cretaceous chalk

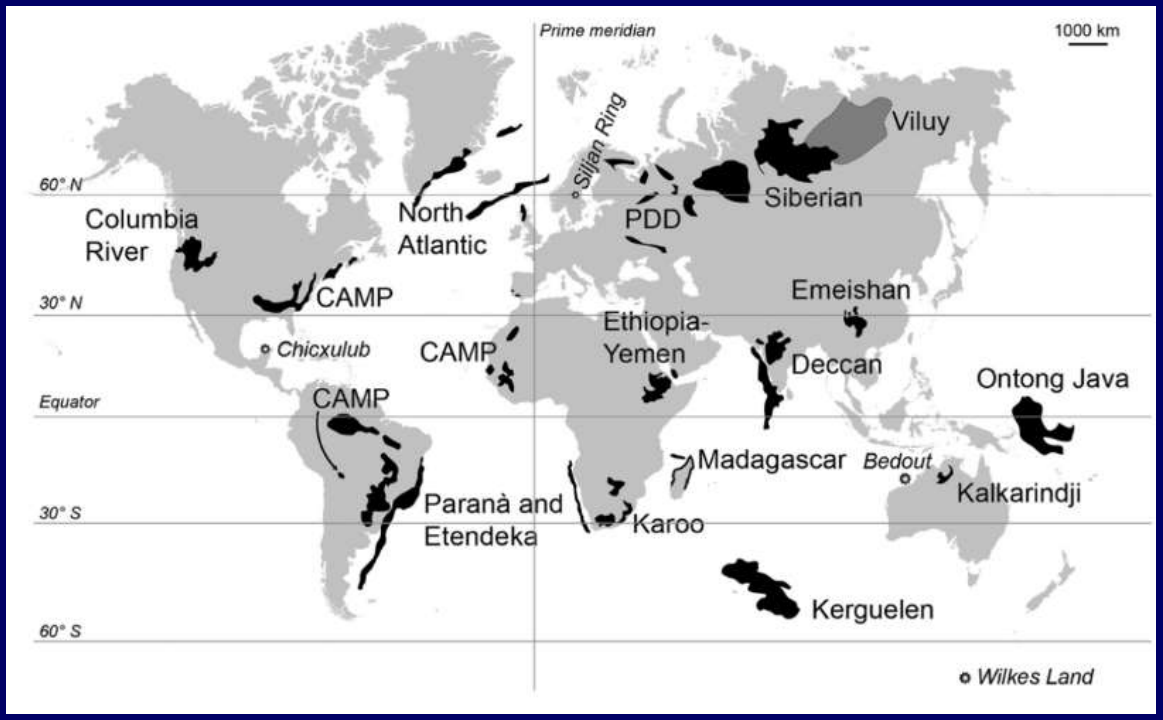


- When the coccolithophore dies, the sphere breaks apart and coccoliths accumulate on the ocean floor
- Coccolith ooze on the ocean floor lithifies to chalk
- Coccolithophores are a major food source for herbivorous plankton whose faecal pellets are deposited in pelagic sediments
- Influence the global carbon cycle because CO_2 is effectively removed from the atmosphere and buried on the sea floor



Not all Large Igneous Provinces (LIPs) = extinctions*





* Particularly in the Cretaceous

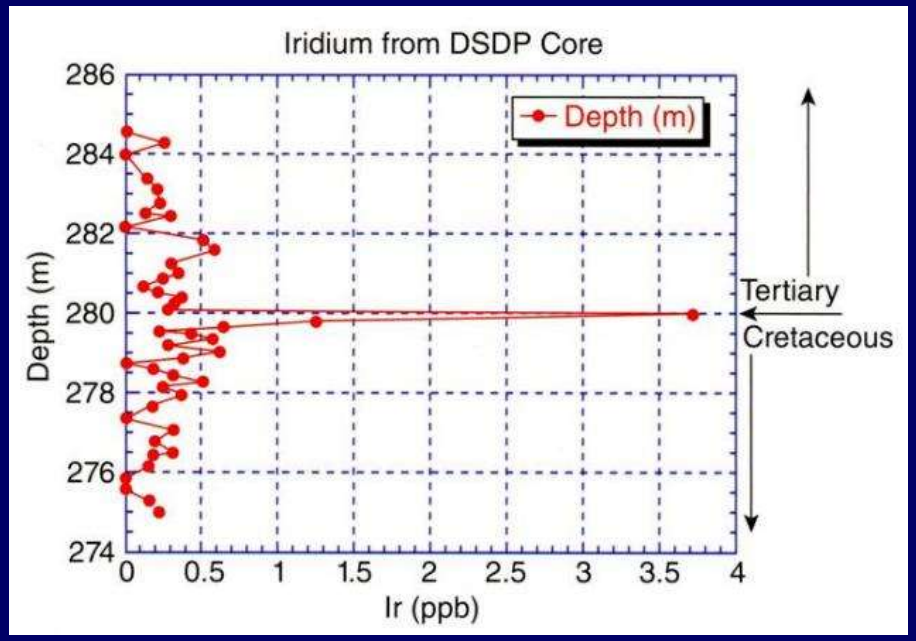
Whether volcanism and environmental changes → biotic crises is function of:

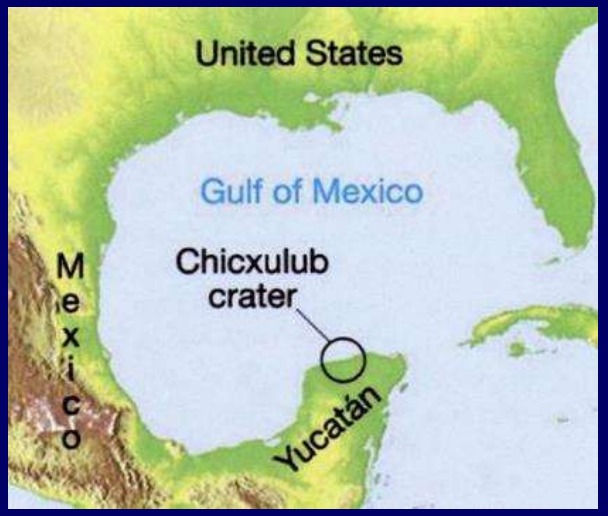
- configuration of continents
- latitude, volume, rate and duration of eruptions
- where volcanism occurs (continent or ocean)
- ability of biota to adapt to environmental change
- pre-existing climate

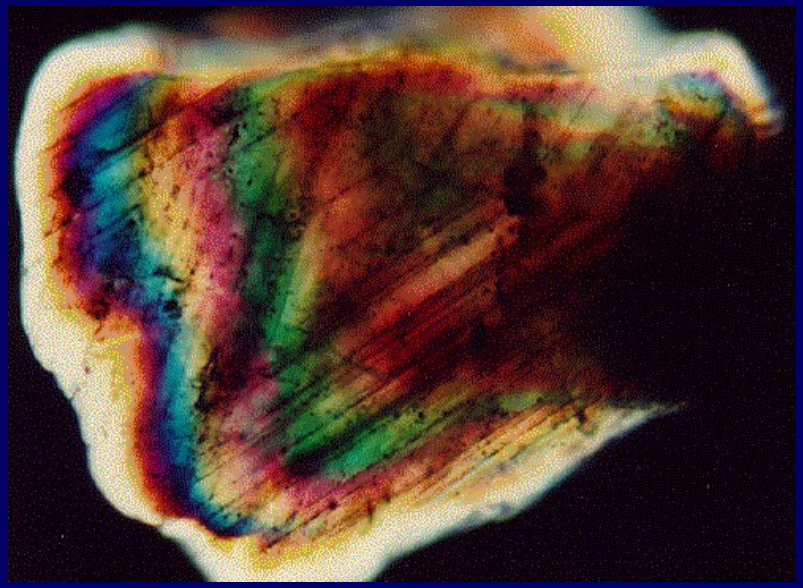
Break



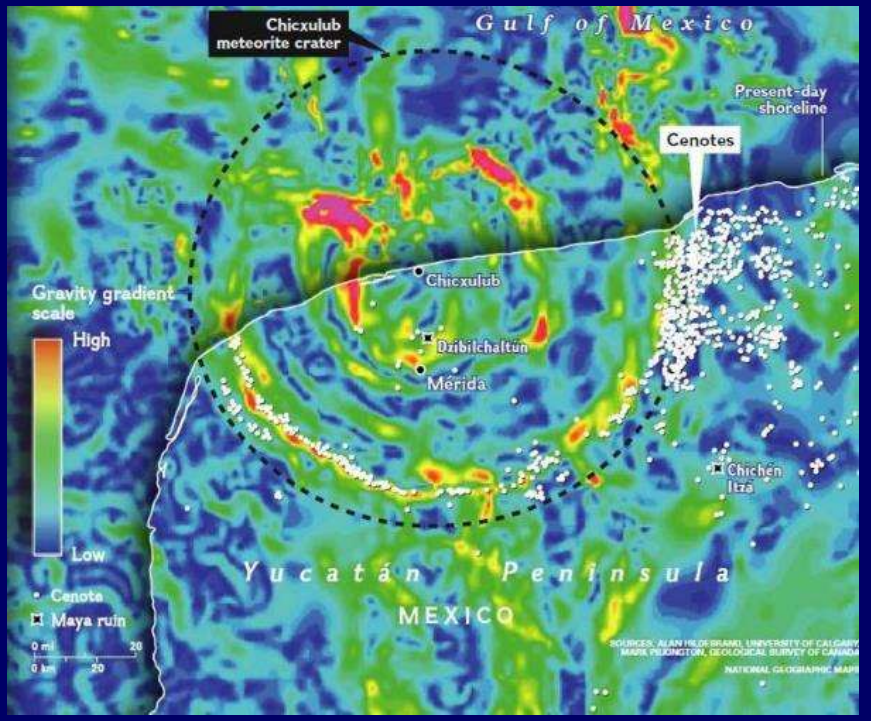
The final act: Causes of the K-T mass extinction





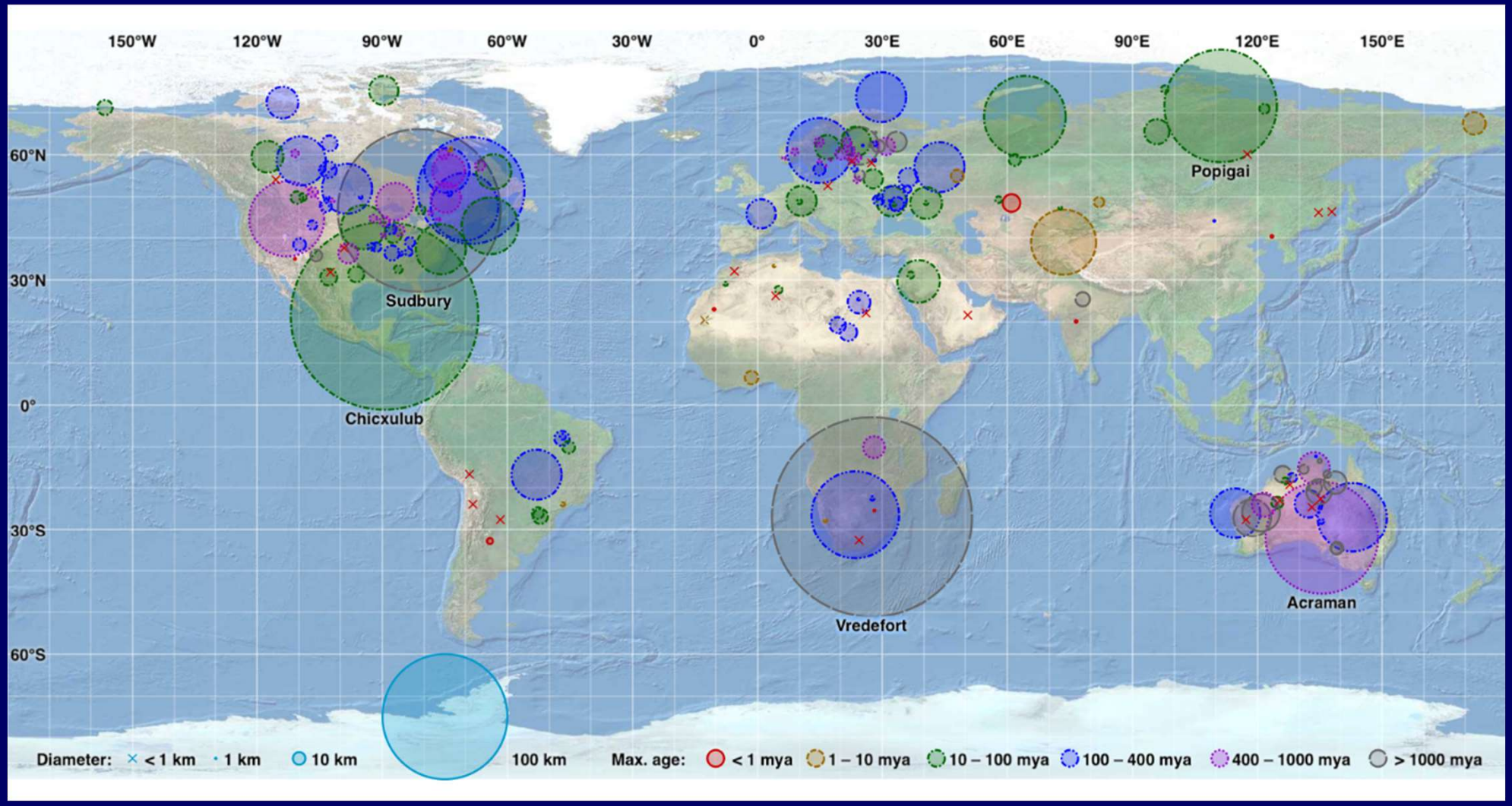


Shock metamorphism of a quartz grain



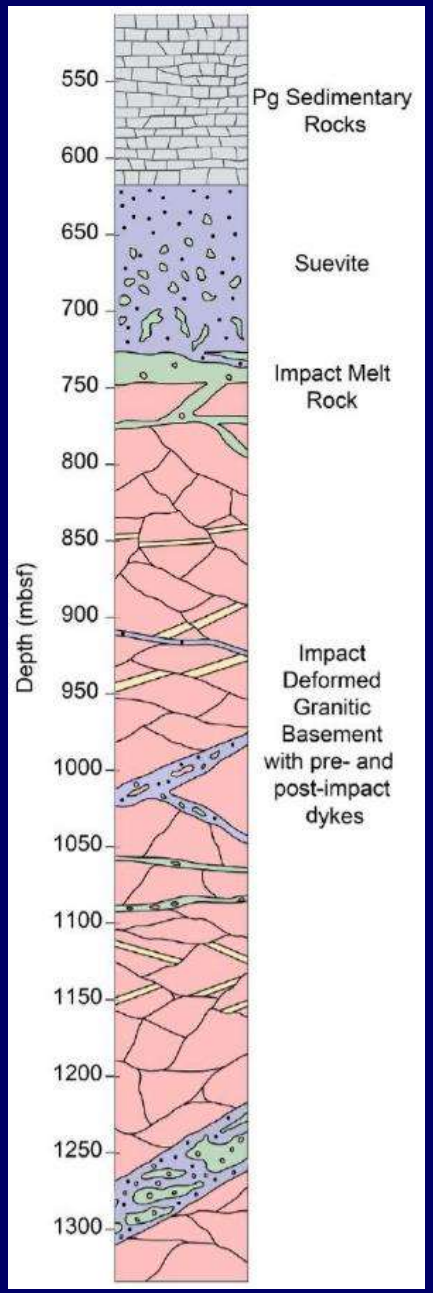
180 Km buried crater formed by 12 km diameter asteroid

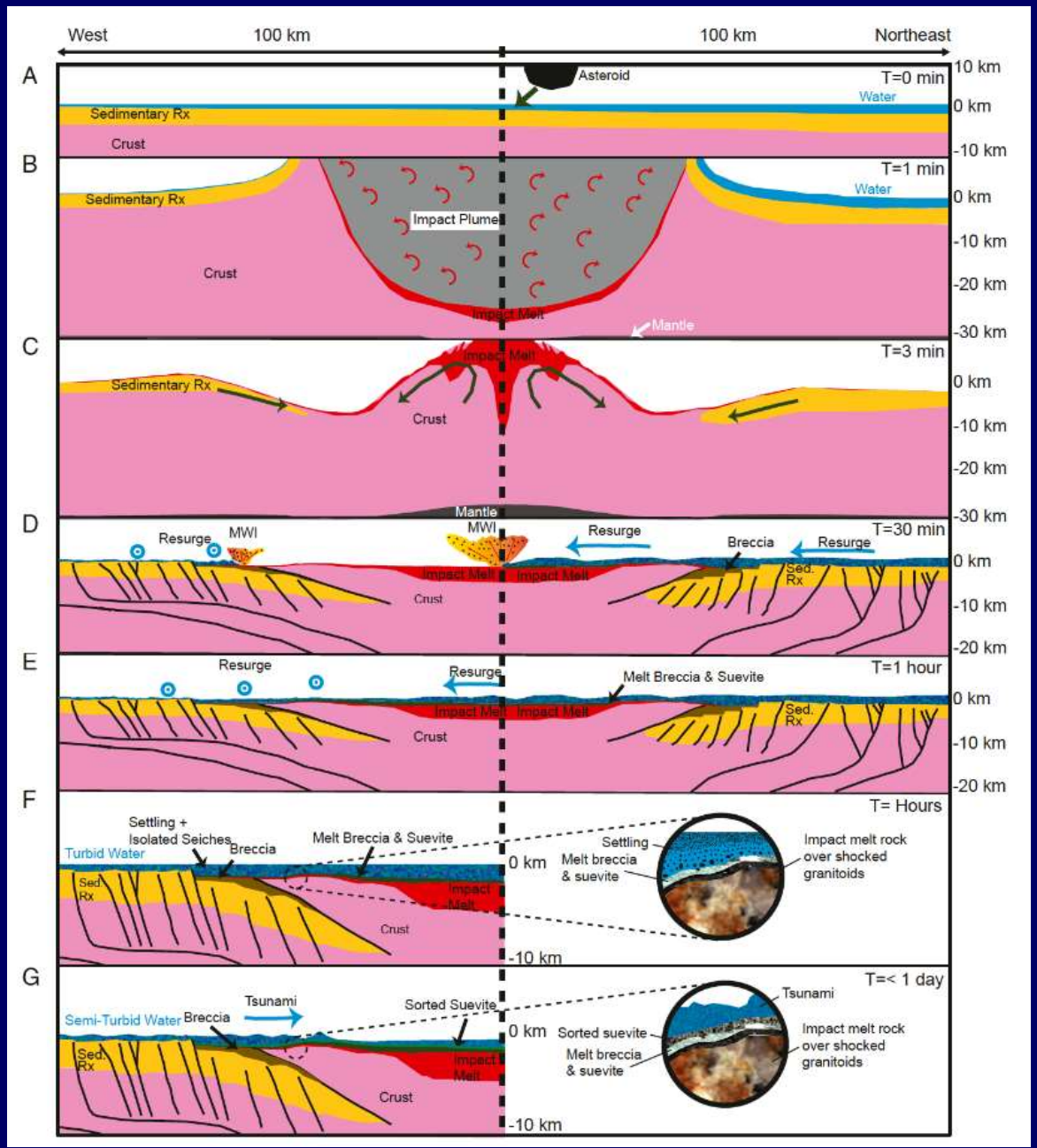
Significant meteorite impacts in Earth history



At least nine large >20 km impact craters formed in the Mesozoic

Immediate consequences of the impact (ground zero)









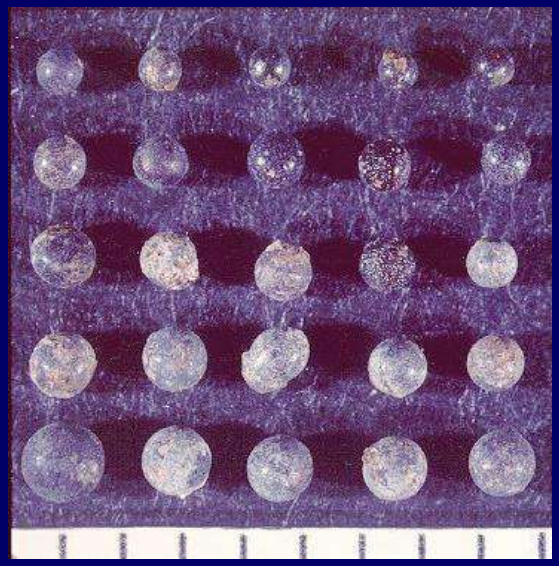




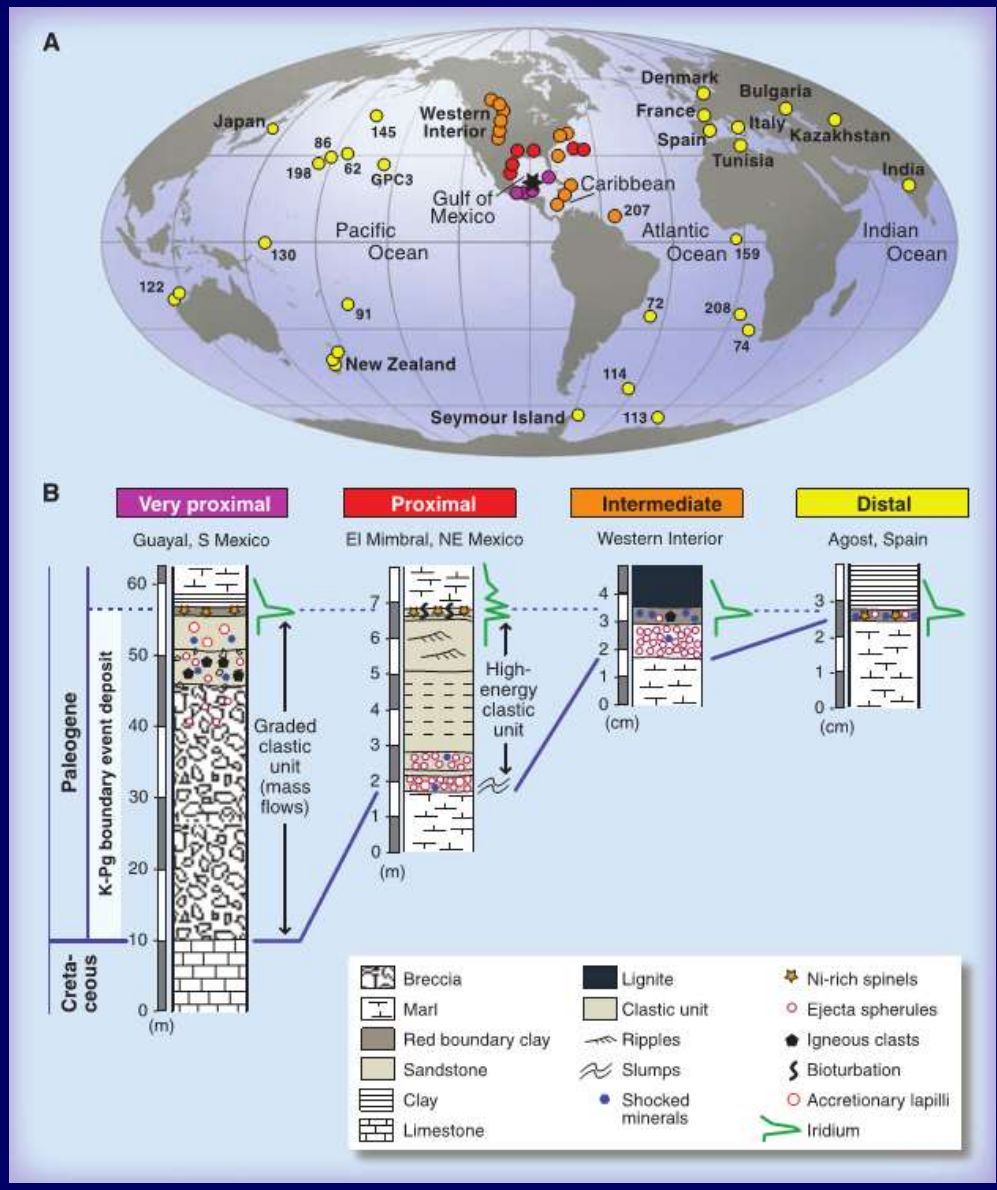
Global consequences of the impact

1. Wildfires triggered by the temperatures of the impact plume and then by atmospheric heating during re-entry of the debris up to 1500 km away

2. Acid rain (nitric acid from shock-heated atmosphere, sulfuric acid from vaporized sulfates, acid rain from wildfires) → crisis for plants and calcareous plankton → crisis up food chain



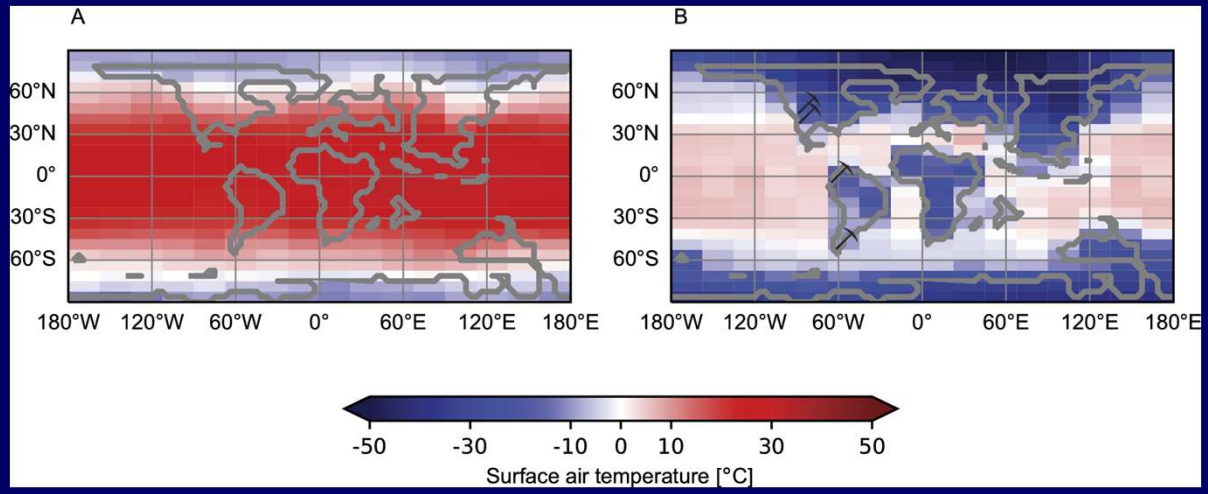
Altered tektites or glass spherules



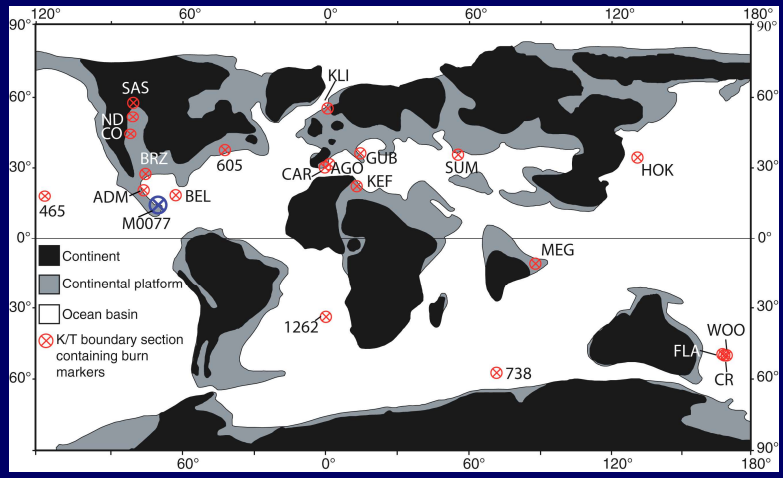
But is the impact the main cause of the K-T extinction?

Global consequences of the impact

3. Injection of sulfur, dust and soot into the atmosphere → lowered global temperatures by over 20°C; disruption of climate ~ 30 yr



Change in global temperature before and after impact



Evidence for burning

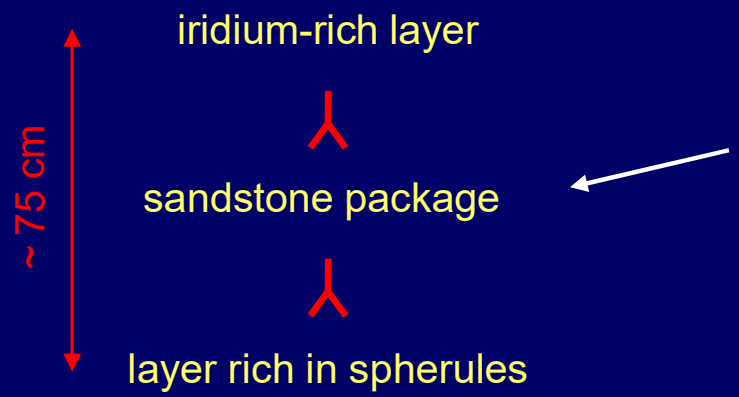
4. Injection of dust and soot into the atmosphere → reduced light → collapse in photosynthetic marine plankton



But is the impact the main cause of the K-T extinction?

The Transitional Unit holds the key

Above the suevite is the Transitional Unit

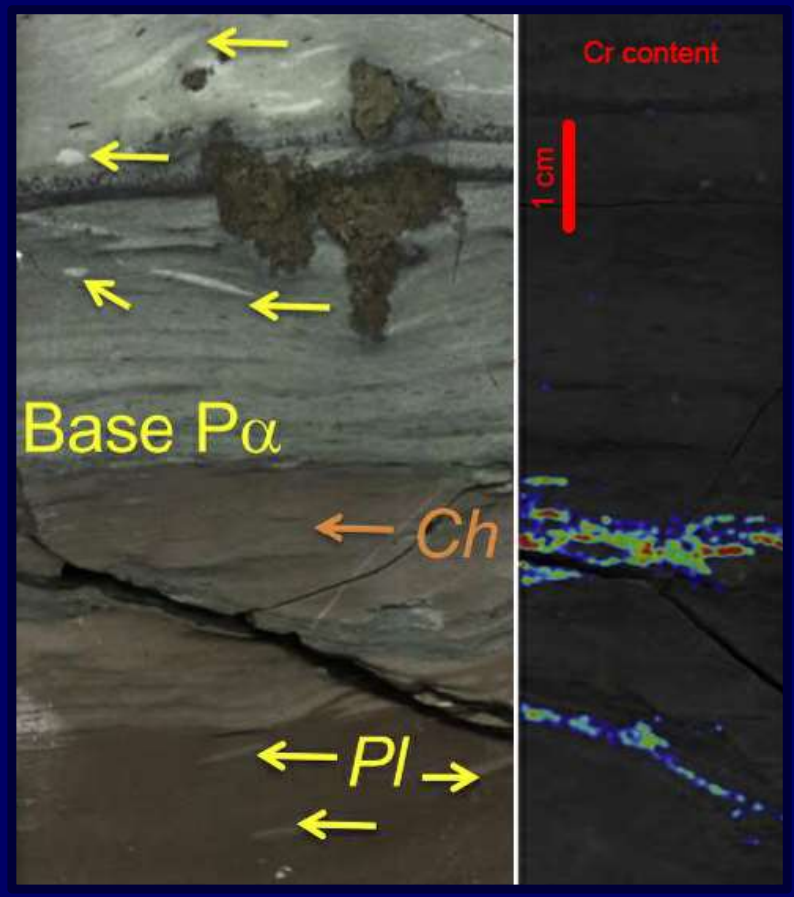


Upward fining from dominantly sandy to dominantly silty with accompanying decrease in size and abundance of impact debris

Thin graded units; thin cross-laminated units with evidence of oscillatory flows

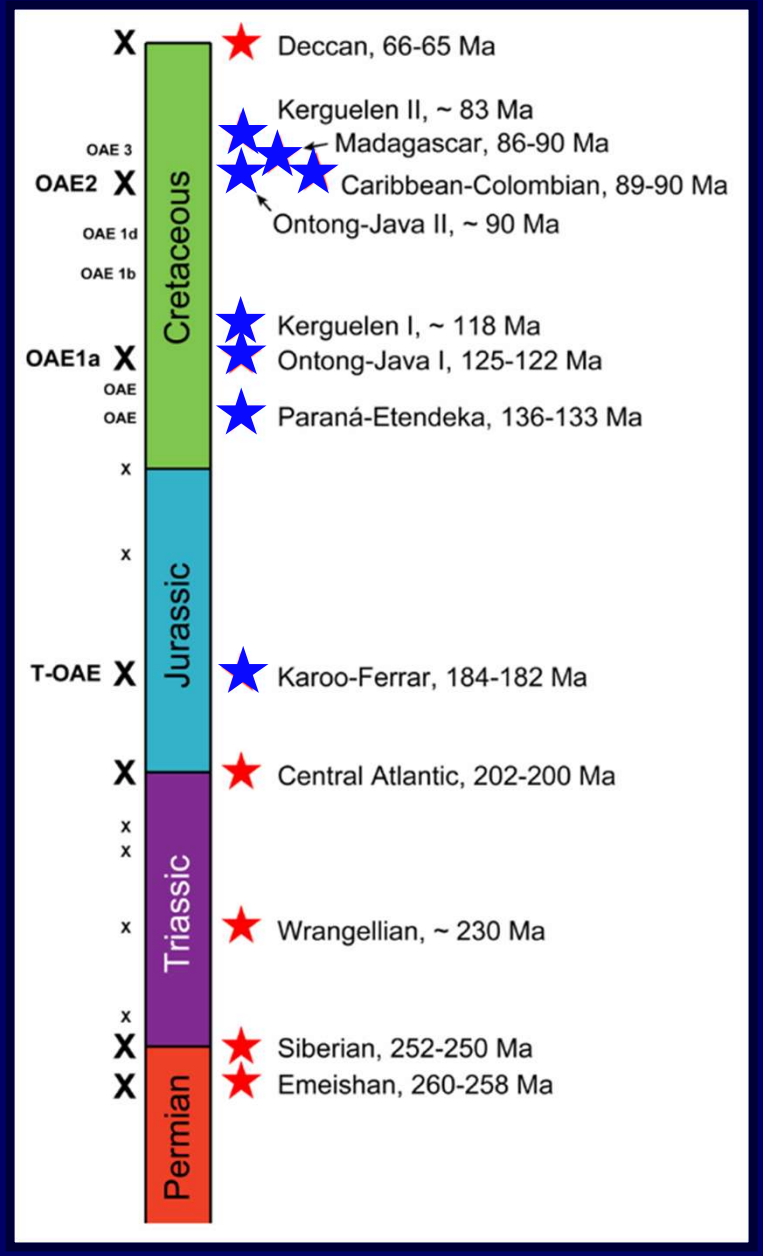
Uppermost part contains tiny burrows (*Chondrites*, *Planolites*)

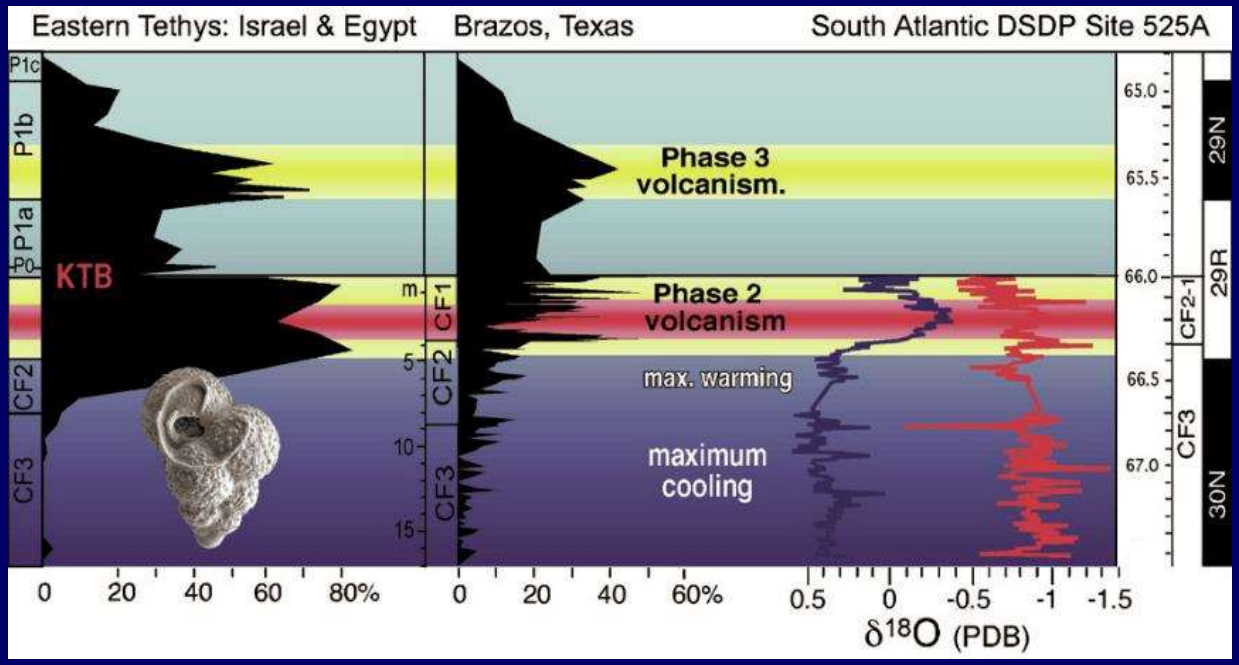




- Ongoing settling of dust through atmosphere and then water column, reworked by gradually waning tsunami / seiches set up by ongoing slope collapse events ~ 6 years
- Return of life to impact site is remarkably rapid

Deccan Traps

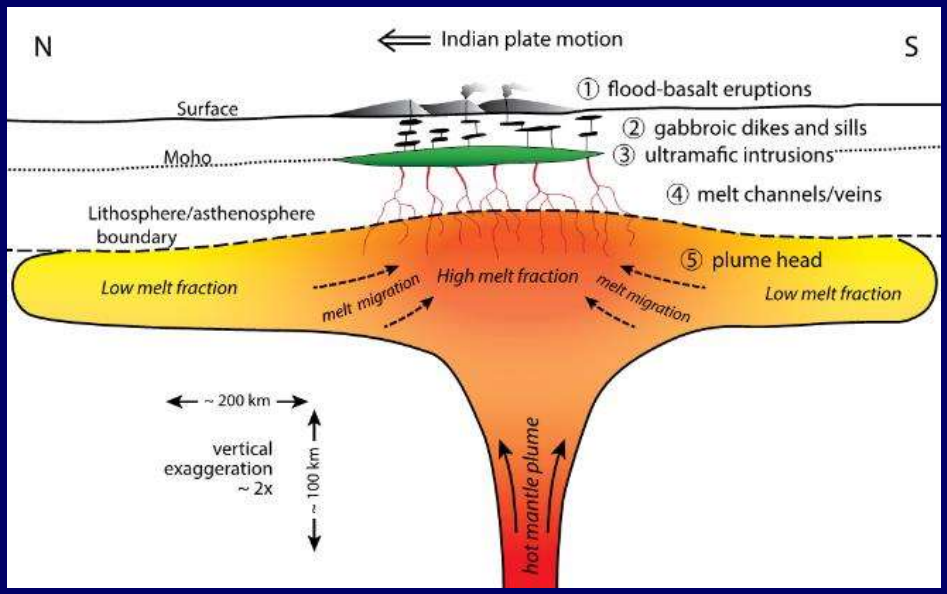


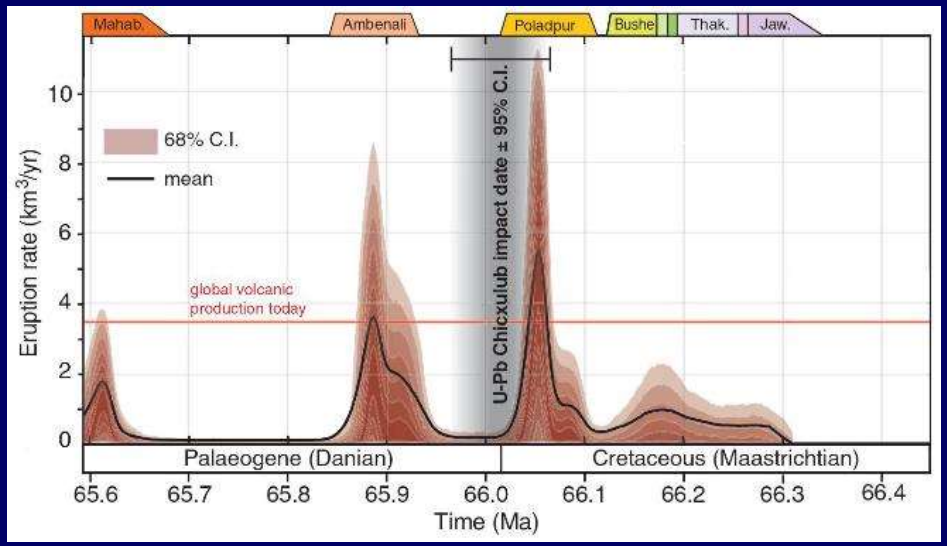




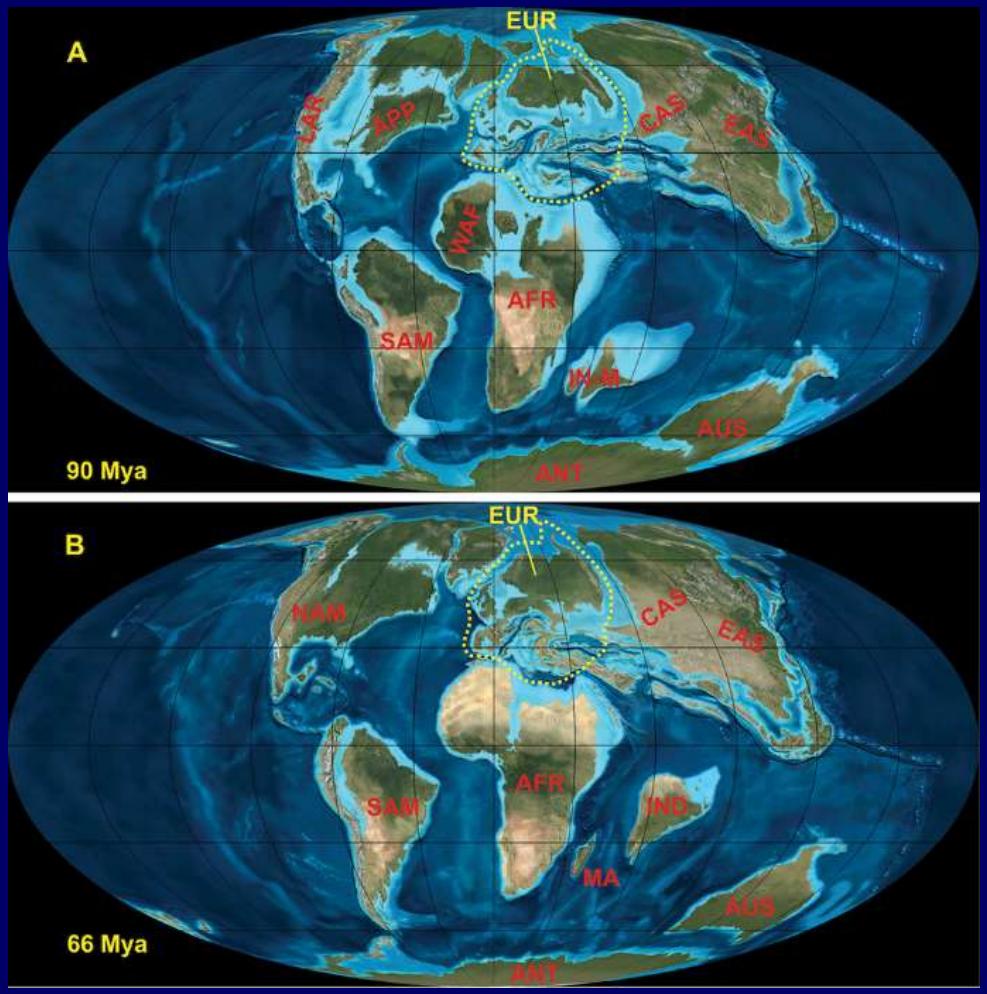
The Deccan Traps (~1.5 million km^3 basalts) were erupted 68-64 Ma, ~80% in an eruptive phase just prior to the K-T boundary

Impact triggered a state shift in Deccan volcanism?





Existing dating evidence points to a change to much more massive and episodic eruptions within ~ 50 kyr after the impact



In the latest Cretaceous there is a background picture of environmental change:

- sea levels were falling → reduced area of continental shelf habitats
- in the last 5 myr of the Cretaceous the global climate was cooling

Mesozoic Era: summary

- Mesozoic era lasted 186 Ma (between 252-66 Ma) and comprises the Triassic, Jurassic and Cretaceous periods
- The supercontinent Pangea rifted into separate land masses leading to sea level transgression and episodic volcanism (LIPs) resulting in major climatic changes
- Cool greenhouse conditions alternated with warm greenhouse and even hothouse conditions, global average temperatures ~6–9 °C warmer than present
- Global sea levels were mainly high, and the content of atmospheric O_2 was varying between 15 and 25%
- Degassing from volcanism linked to the rifting process of Pangaea increased atmospheric CO_2 levels by up to 16x higher than present, exerting a major climate control
- Continental break-up modified landmass configurations generating huge epicontinental seas and altering oceanic circulation patterns
- Mesozoic extinction events have been related to the effects of rifting and flood volcanism and/or asteroid impacts (K-Pg)

Next week

The Cenozoic Era: The modern Earth arrives

The Great Tertiary cool down

Causes of Pleistocene ice age and its oscillations

The record in ice cores

Climate variability in the late Holocene

Last Glacial Maximum

The Younger Dryas: a signpost for the oceanic role of in climate

Other Cretaceous pelagic sedimentary rocks: Cherts



Key characteristics:

- very hard
- finely crystalline quartz
- formed by solution precipitation
- often slightly glassy in appearance
- conchoidal fracture



Well bedded primary chert



Secondary chert e.g., flint nodules in chalk

Organisms whose hard parts are made of silica

- Organisms who secrete exoskeletons as amorphous silica = opaline silica $SiO_2.nH_2O$
- Two main groups – diatoms (Cretaceous onwards) and radiolaria (early Cambrian onwards)
- Diatoms – phytoplankton (floating photosynthetic algae) – mainly freshwater, some high latitude marine
- Radiolaria – zooplankton (microscopic animals with planktonic lifestyle), common in equatorial seas





Precambrian BIFs: alternating chert and iron oxide bands
This chert ppt directly from supersaturated sea water, not organisms



Formation of siliceous oozes and primary cherts



Typical sequence of radiolarian cherts

Laminae in siliceous oozes reflect seasonal variation in plankton productivity and/or seasonal influxes of detrital clay

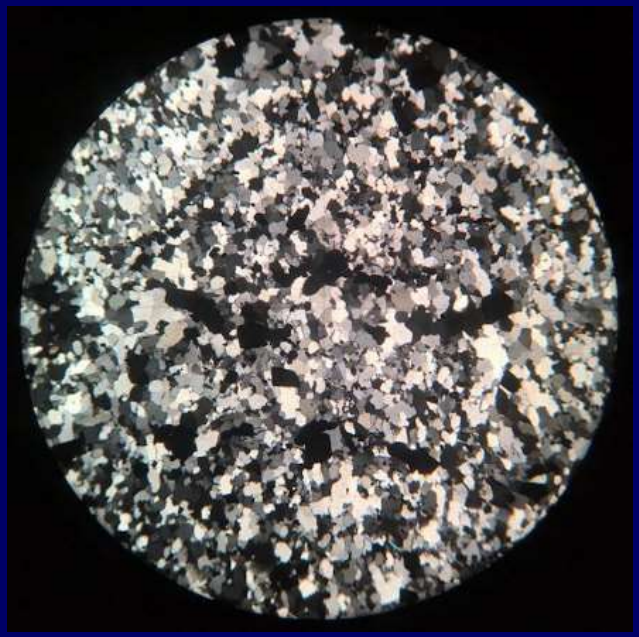
Why is it red?

Radiolaria and diatoms live in surface waters of ocean and sink to sea floor after death

Once incorporated in sediment and buried, the radiolaria/diatoms will slowly re-crystallize to quartz

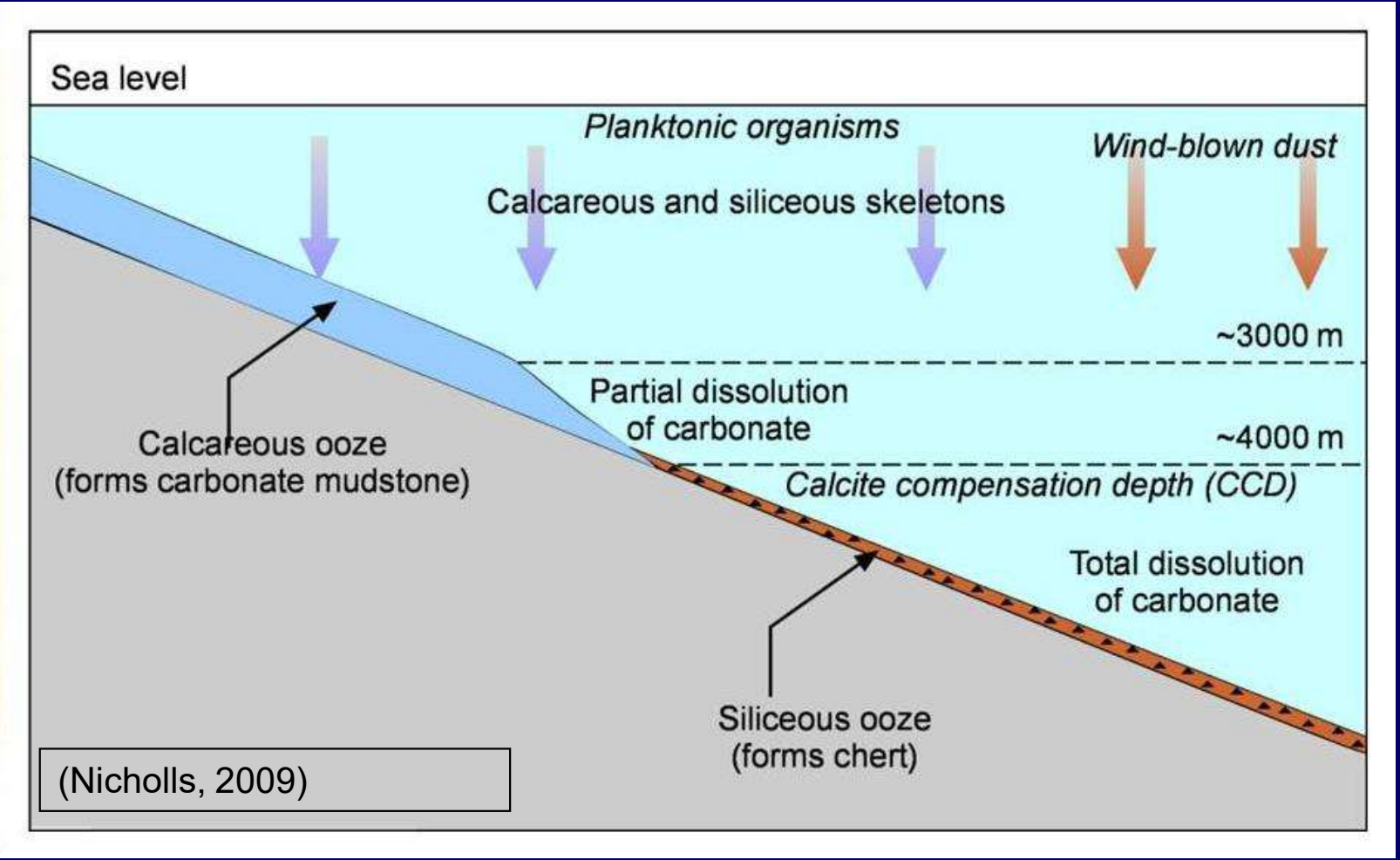
This takes a long time, typically tens of millions of years

Eventual product is a bedded chert



Thin section of cherts: note you can't see the individual radiolaria or diatoms as they have been recrystallised

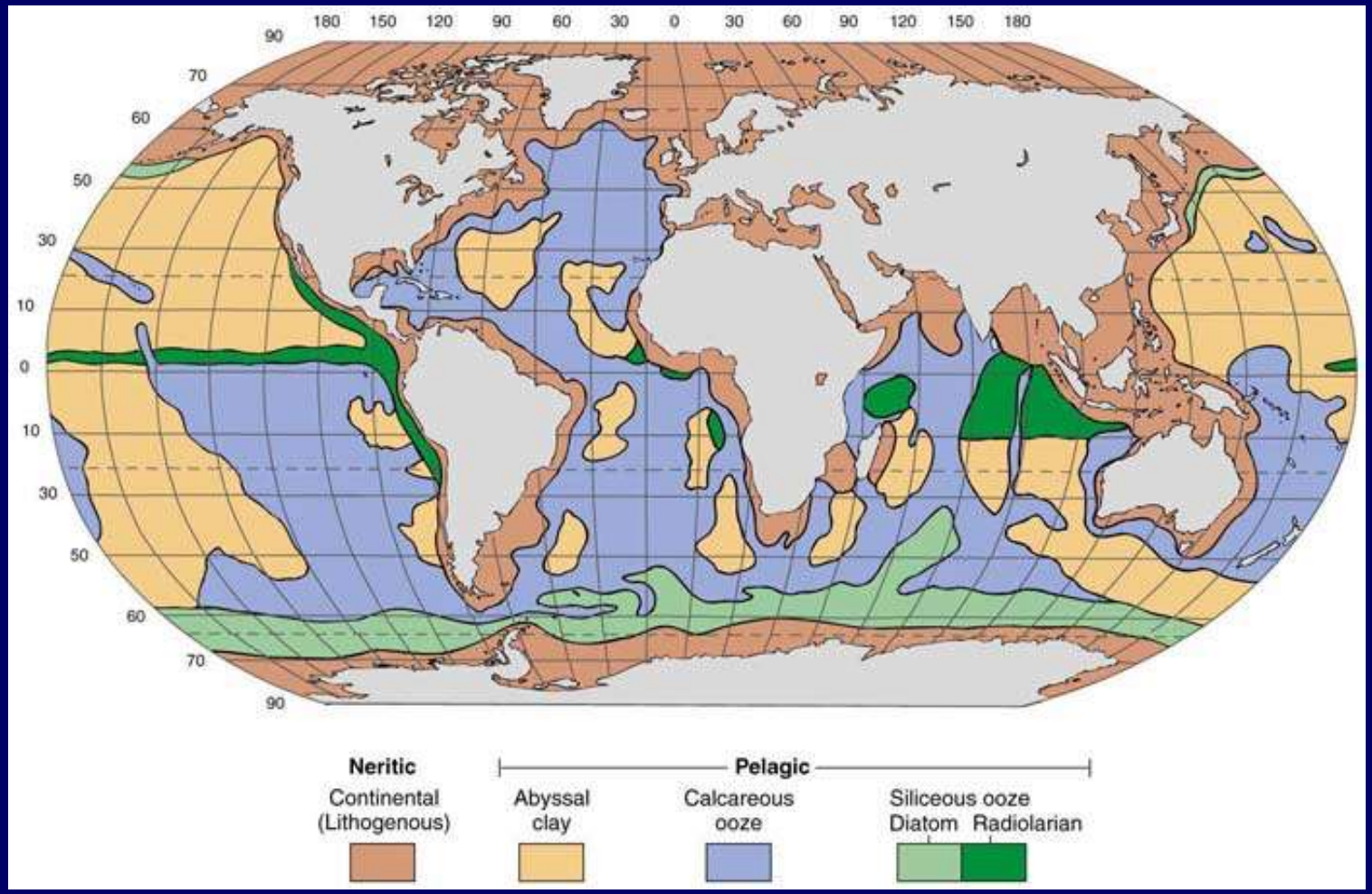
Calcite compensation depth



The CCD controls the distribution of siliceous and calcareous oozes on the ocean floor

- $CaCO_3$ solubility increases with depth (due to temperature decrease and pressure increase)
- At CCD rate of calcium carbonate material forming and sinking is equal to the rate the material is dissolving
- Most opaline silica dissolves in surface waters (warmer, Al/Si of shells, high surface area) only about 3% of global silica flux is buried in sediment

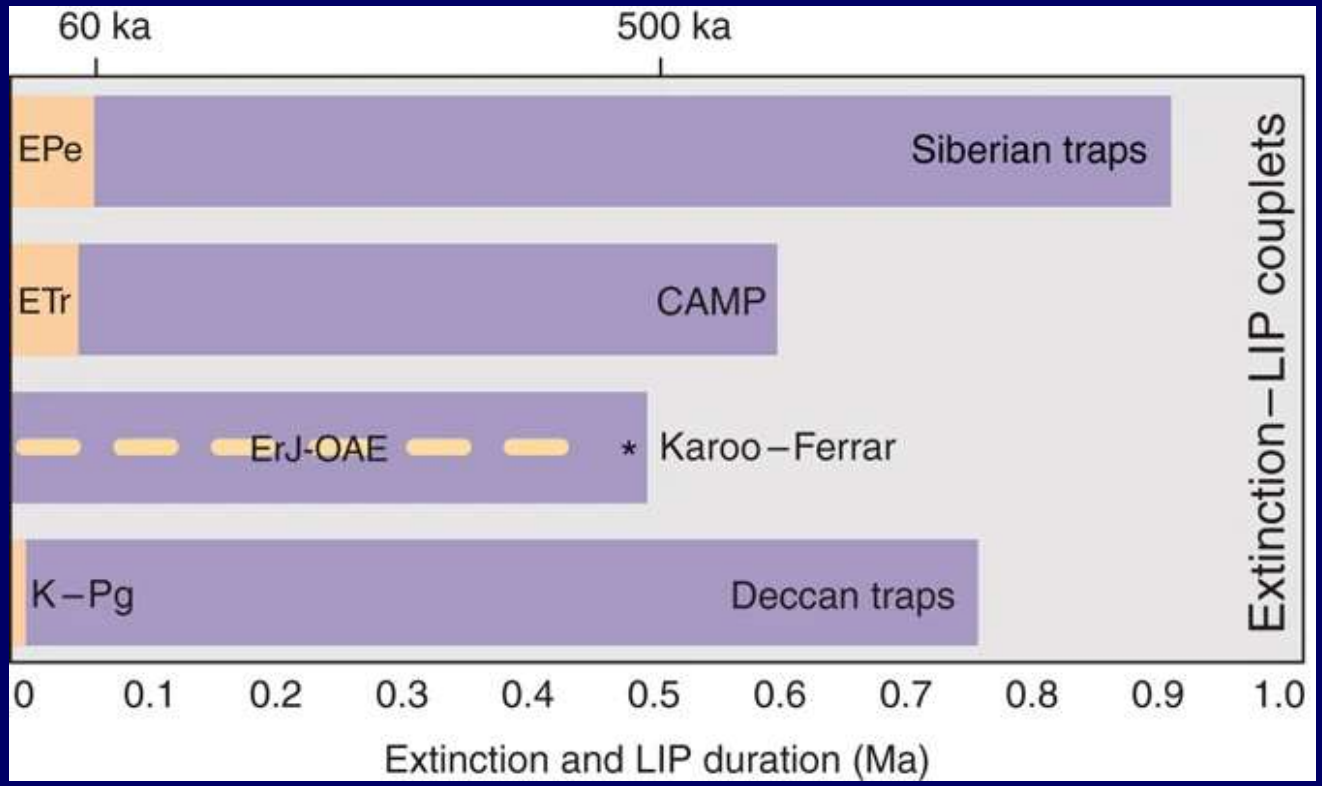
The depth of the CCD has changed through time



Distribution of pelagic and neritic sediments today

- Cretaceous - early Eocene the CCD was much shallower due to intense volcanic activity = higher atmospheric CO_2 concentrations (higher pCO_2 in oceans) → increased dissolved CO_2 in surface waters
- Palaeozoic and early Mesozoic, prior to evolution of coccolithophores, siliceous oozes may have deposited at much shallower depths than when there was calcareous plankton available for sedimentation

The case against volcanic causes for Mesozoic extinction events



End-Permian

End-Triassic

Early Jurassic Ocean Anoxia Event

K-T or K-Pg

1. Significant disparity in the timescales over which LIP magmatism and mass extinction occurs:
   - magmatism lasts about 1–5 Ma whereas mass extinction happens on timescales of <100 kyr
2. Relative timing of LIP emplacement and mass extinction.
   - rather than both events coinciding at onset, LIP emplacement may have been 100ka's prior to mass extinction, with little evidence for negative feedback in the biosphere during voluminous eruptions
3. Not all LIP events are associated with marked environmental change